

The OpenAI Navier–Stokes and Euler Blow-Up Proofs: A Physical Reading, Not a Physical Refutation

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September 2026
Version 5.7.0

Abstract

In September 2026, an OpenAI multi-agent system produced a Lean 4-verified proof of finite-time blow-up for the forced 3D Navier–Stokes equations (Millennium Prize Alternatives C and D) and for the unforced Euler equations. The proof is syntactically flawless: zero **sorrys**, zero custom axioms, no weakened norms. The Clay Millennium Prize problems, Navier–Stokes existence and smoothness included, are precise questions about a specific mathematical model; they were never framed, and are not understood by working mathematicians, as claims about how real fluids behave — the model is already known to break down under compressibility, cavitation, molecular discreteness, and other effects well before any putative singularity. Read that way, OpenAI’s result is exactly what it claims to be: a rigorous statement about the continuum incompressible Navier–Stokes system on $\mathbb{R}^3$.

Our contribution is not a refutation but a physical reading. Because the collapsing core keeps a radial Reynolds number of order one, its scales are diffusive, $\ell_r \simeq \sqrt{\nu t}$ and $u \simeq \sqrt{\nu/t}$ up to factors $t^{\pm h}$ with $h < 1/100$, so the reading is essentially independent of the initial vortex size and of the choice of units. Compressibility, rarefaction and viscous heating then become order-one effects at a single length $\ell_* = \nu/c_s$ (0.7 nm in water, 45 nm in air), reached a few picoseconds (water) or nanoseconds (air) before the mathematical singularity: the three validity limits are one limit, and it can be written as a local vorticity bound $|\omega| \lesssim c_s^2/\nu$. For liquids at ambient pressure a fourth limit, cavitation, is reached three decades earlier, when the core is still 70–130 nm wide. We also show that (i) intensive local energy density and enstrophy diverge ($\sim \tau^{-1.010}$, $\sim \tau^{-0.515}$) while the Prize’s global L^2 bound is respected; (ii) the apparent 10^{28} condition number of the proof’s moment-matching system is a non-dimensionalization artifact ($\kappa \approx 4.1 \times 10^5$ once scaled); (iii) the force is specified by working backward from the singularity, a legitimate existence-proof technique that leaves it without independent physical origin; and (iv) the Euler datum requires coherent oscillatory structure at scales below any molecular cutoff, where the missing physics is viscosity itself. None of this makes the Lean proofs wrong, and none of it proves anything about turbulence in real fluids — both points made forcefully by the fluid-dynamics community when an early version of this audit was shared publicly (§11). Finally, we test the cutoff-regularization hypothesis that earlier versions of this paper could only state: it is exact given its premise, but that premise — a $\text{Re} \approx 1$ viscously dominated core — is not produced by generic initial data, so the arrest scale instead obeys inertial-range scaling, and the hypothesis cannot be tested without reproducing the construction’s own forcing. We then supply that forcing. On a manufactured $\text{Re} \approx 1$ collapse the barrier’s engagement reduces to a function of a single variable, while transport-filtering regularizations of the Leray- α and LANS- α type are weaker by one to two orders of magnitude, because on such a collapse they can act only through a nonlinearity that is a few percent of the dynamics. Nor is the missing lock a stronger drain: the exact shear-mode spectrum of the BGK kinetic model damps small scales *less* than viscosity, caps the relaxation rate at the collision frequency, and ends the hydrodynamic mode at $k\lambda = \sqrt{\pi/2}$; a nonlinear discrete-velocity BGK simulation of the

same forced collapse confirms that the kinetic model does not arrest it there. What operates where the continuum ends is not a regularization of the hydrodynamic equations but the termination of the hydrodynamic description itself — on the route this construction takes. Rereading Tao’s averaged-equation blow-up and a recent human-written Euler blow-up through the relation $\text{Kn} = \text{Ma}/\text{Re}$ shows that the three scenarios meet different physics first: kinetic termination, compressibility and heat inside the continuum, and plain viscosity. On the second, a one-dimensional compressible Navier–Stokes–Fourier simulation finds that air does not follow a driven swirl past a local Mach number of 0.70; on the first, neither compressibility nor heat stops the core before ℓ_* . A molecular-dynamics simulation with no continuum closure reproduces the lock where the gas remains a continuum, and shows the core emptying into a free-molecular region beyond it; in a liquid it finds cavitation, after which the swirl at the cavity wall stays within the hollow-vortex bound at the ambient pressure measured in the run; a water-like liquid first exceeds the bound stated at the initial pressure, a registered prediction that failed, because the closed box’s ambient pressure rose. The claim that a Leray- α filter of width ℓ_* represents the fluid at its continuum limit is withdrawn, with a formal reason: the filter width does not appear in the linearized dynamics. On the formal side, we prove in Lean 4, on OpenAI’s own definitions and with no additional hypothesis, that any object with their candidate properties exceeds every velocity-gradient bound arbitrarily close to the singular time.

1 Introduction

The application of large-scale AI agents to formalize mathematics in Lean 4 guarantees logical consistency: if the proof compiles, the chain $A \implies B$ is syntactically flawless. On September 8, 2026, OpenAI announced that an internal multi-agent system had produced such a proof of finite-time blow-up for the forced 3D Navier–Stokes equations and, separately, for the unforced Euler equations [2, 4, 3]. (Figures for the size of that system — “10,000 agents” — circulated in secondary coverage but appear in neither paper; we do not repeat them as fact.) The announcement was accompanied by a public discussion of priority with other groups working on the Euler problem. We do not adjudicate it, and it plays no role in what follows, except as a reminder that a result produced under competitive pressure, however impeccably kernel-checked, still benefits from independent, unhurried scrutiny of what it actually establishes — which is the purpose of this paper.

The result does not arrive in a vacuum. Finite-time blow-up was known for an averaged Navier–Stokes model [18], for $C^{1,\alpha}$ Euler data [31], and, with computer assistance, for smooth Euler and Boussinesq data in domains with boundary [32] following numerical evidence [30]; non-uniqueness of Leray solutions of the *forced* Navier–Stokes equations [34] had already shown how much a well-chosen force can do. What is new is smooth, compactly supported data on $\mathbb{R}^3$ (Euler) and a smooth force acting on a fluid at rest (Navier–Stokes), with kernel-checked proofs.

Our purpose is not to contest the mathematics. A proof mapping $A \implies B$ is syntactically true regardless of how physically realistic A is, and the Millennium Prize problems were never framed as claims about physical fluids: Fefferman’s statement [1] asks precise questions about a specific continuum model, chosen because it is tractable and physically motivated, not because it is physically exact. This is also the reading that emerged in an open discussion of an early version of this work among fluid dynamicists on the `r/FluidMechanics` forum [6] (§11): that Navier–Stokes has always been a useful approximation rather than an exact description, that a blow-up result mainly locates where a specific mathematical model runs out of validity, and that the more interesting question is what a genuinely physical admissibility criterion — analogous to the entropy condition that selects physical weak solutions of the inviscid Burgers equation — might look like.

Since the Lean 4 kernel forecloses algebraic error, this paper’s contribution is not a bug hunt in that sense. It is instead a quantitative account of how far OpenAI’s specific constructed solution travels from the physical regime the Navier–Stokes model is normally used to describe, built from

scaling analysis, the dimensional structure of the proof’s own lemmas, and constitutive physics (§§2–8). Section 9 then tests what, if anything, stops such a collapse: a cutoff regularization and its premise, a manufactured $\text{Re} \approx 1$ forced core, transport filters against dissipation, kinetic theory (linear and nonlinear), and compressibility and heat; it closes with a regime map that places this construction, Tao’s averaged-equation blow-up and a forced Euler blow-up on one diagram, and with the withdrawal of our earlier identification of a Leray- α filter width with the continuum limit. An outlook (§11) sets out research directions and an architecture for grounding AI proof search in physical admissibility, including Lean 4 statements proved on OpenAI’s own definitions. A note on how this version differs from earlier public versions is given in Appendix A.

2 The Syntactic Health: Lean 4 Purity and Gevrey Cutoffs

2.1 AST Verification Results

An automated crawl of the Abstract Syntax Tree confirms the proof is structurally flawless:

Criterion	Expected	Found
<code>sorry</code> / <code>admit</code> in core proof	0	0
Custom axiom declarations	0	0
<code>opaque</code> / <code>constant</code> declarations	0	0
Force smoothness type	<code>ContDiff ℝ ∞</code>	<code>ContDiff ℝ ∞</code>
Force support	Compact Positive Time	Compact Positive Time
Energy bound	$L_t^\infty L_x^2$ (uniform)	$\exists E, \forall t, \text{kineticEnergy } u \ t \leq E$
Sobolev weakening detected	None	None

Table 1: Lean 4 formal verification results. `sorry` instances exist in `ComparatorChallenges/`, a separate module containing intentional open problems—not part of the core proof.

There are no opaque axioms, no weakened Sobolev norms, and the global L^2 kinetic energy is uniformly bounded: the formalization proves the statement the Prize asks for.

2.2 The Gevrey Bypass: Satisfying C^∞ Without Real-Analyticity

The Millennium Prize requires a perfectly smooth (C^∞) external force. Were the force required to be real-analytic (C^ω), the construction would be impossible outright: a function that is real-analytic and vanishes outside a bounded region must be identically zero everywhere (the identity theorem), so no nontrivial compactly supported force could exist at all. The formalization sidesteps this rigidity with Gevrey-class cutoffs of the form $\chi(q) \sim \exp(-1/q^2)$, which are C^∞ but not real-analytic.

By classical asymptotic analysis, $\exp(-1/q^k)$ has Gevrey index $s = 1 + 1/k$; for $k = 2$ this gives $s = 1.5$ (finite-sample `SymPy` fits with $N \leq 15$ show an empirical pre-asymptotic slope near 2.22, reflecting the fact that the true asymptotic regime is only reached for larger N). This specific cutoff therefore lies in $G^{1.5} \setminus C^\omega$: infinitely differentiable, with all flat limits $\lim_{q \rightarrow 0^+} d^N[q^{-A} \exp(-1/q^2)]/dq^N = 0$ holding for every N and A (confirmed symbolically), yet not real-analytic. That gap between C^∞ and C^ω is exactly what a compactly supported, nontrivial smooth force requires — there is nothing improper or exotic about invoking it; every physically standard “bump function” construction relies on the same gap. What is worth flagging is not that the cutoff is Gevrey rather than analytic (that is necessary, not suspicious), but how

the resulting velocity profile behaves physically once combined with the rest of the construction, which is the subject of §4–§7.

2.3 The Proof Architecture: Internal Reynolds-Stress Cancellation

The proof’s logical structure is easily mistaken for a simple Method of Manufactured Solutions (MMS). In standard MMS, one merely picks an arbitrary smooth velocity field u and *defines* $f = \partial_t u + (u \cdot \nabla)u - \nu \Delta u + \nabla p$, which is automatically smooth because u was smooth to begin with.

The OpenAI proof is categorically more sophisticated. The paper itself acknowledges [2]:

“For any incompressible flow u and pressure p , we can always define the external force f to be the residual in (1.1). The Navier-Stokes equations then hold by construction.”

But the constructed u is a singular collapsing vortex column. Its raw residual is *not* smooth— $f_{\text{raw}} \rightarrow \infty$ as $\tau \rightarrow 0$. The proof’s core intellectual contribution, spanning Sections 6–9 of [2], is the construction of oscillatory wave packets whose *internal nonlinear Reynolds stresses* $-\nabla \cdot (u' \otimes u')$ precisely cancel the divergent residual of the collapsing core. It is these *internal* stress interactions—not the external force—that tame the singularity. Only after this Reynolds-stress cancellation homogenizes the residual is the remaining smooth remainder declared as $f \in C_c^\infty$.

The Lean 4 code makes the final assignment explicit. In `CandidateFromLimits.lean` the force is defined as the smooth extension, past $t = 1$, of the traced residual of the constructed velocity and pressure, with the docstring “No force is an input to this definition”; the theorem `force_eq_activated_residual` states that on $0 \leq t < 1$ the force equals the Navier–Stokes residual of the final (u, p) . The force does not *drive* the singularity; it is what is *left over* once the internal wave dynamics have made that residual smooth.

3 Matrix Scaling & Non-Dimensionalization of the Moment System

To seamlessly stitch the singular inner vortex to the smooth outer heat flow, the formalization matches five cumulative radial moments via a 5×5 linear system, which we write with matrix A (Proposition B.8 and the moment conditions (5.10)–(5.11) of [2]). A naive SVD of this unscaled 5×5 dimensional matrix yields a condition number scaling as $\kappa(A) \sim X_R^{7.75}$, which exceeds 10^{28} as $X_R \rightarrow 1000$.

However, rigorous non-dimensionalization analysis reveals that this condition number explosion is a classical numerical linear algebra artifact. The entries of A integrate polynomial basis functions $(x \cdot X_R)^{2i}$ and $(x \cdot X_R)^{2j}$, spanning physical dimensions from X_R^0 up to X_R^8 . Mathematically, the matrix factors as:

$$A = D B D, \tag{1}$$

where $D = \text{diag}(1, X_R^2, X_R^4)$ for the azimuthal block (and $D_z = \text{diag}(X_R, X_R^3)$ for the axial block), and B is a non-dimensional matrix composed of integrals of dimensionless coordinate $x^{2(i+j)}$.

Applying the diagonal preconditioning matrix D^{-1} removes the X_R dependence entirely. The non-dimensionalized condition number $\kappa(B) \approx 4.11 \times 10^5$ is well-conditioned and uniform for all X_R . Consequently, the moment-matching system is not a source of fragility. Two further remarks are in order. Values of κ above $\sim 10^{16}$ computed in double precision are not reliable in any case; and the condition number of a linear system used to *choose parameters in a proof* is not a dynamical instability of the flow, and should not be read as a measure of fine-tuning. The rigorous version of “how fine-tuned is this singularity” is the codimension question of §11.

X_R	$\kappa(A)$ (Raw Unscaled)	$\kappa(B)$ (Non-Dimensionalized)
1	4.11×10^5	4.11×10^5
10	2.36×10^{12}	4.11×10^5
100	2.36×10^{20}	4.11×10^5
1000	1.78×10^{28}	4.11×10^5

Table 2: Condition number comparison of raw dimensional matrix A versus non-dimensionalized matrix $B = D^{-1}AD^{-1}$. Once properly scaled, $\kappa(B) \approx 4.11 \times 10^5$ remains completely bounded and independent of X_R .

4 The Thermodynamic Picture: Divergent Local Quantities

Alternative C of the Millennium Prize requires only that total kinetic energy remains uniformly bounded. The authors achieve this by ensuring the volume of the collapsing vortex core shrinks faster than the velocity-squared diverges. This is a scaling feature, not a thermodynamic one, and it is exactly what the Prize permits.

With velocity scales $u_\theta, u_z \sim \tau^{-1/2-h}$ ($h < 1/100$), radial scale $\ell_r \sim \tau^{1/2}$, and axial scale $\ell_z \sim \tau^{1/2-h}$, a SymPy asymptotic analysis yields the scaling exponents in Table 3. Here τ denotes the dimensionless self-similar time-to-blow-up parameter used throughout the OpenAI construction (not, in this table, a value in seconds — see §5 for the physical time conversion).

Quantity	Exponent (τ^x)	Limit	Prize-compatible?
Global kinetic energy E	+0.4850	$\rightarrow 0$	yes
Intensive local energy density e_{local}	−1.0100	$\rightarrow \infty$	—
Enstrophy $\Omega(t)$	−0.5150	$\rightarrow \infty$	—
L^3 norm $\ u\ _3$	−0.0067	$\rightarrow \infty$	—
$H^{3/2}$ Sobolev norm $\ u\ _{H^{3/2}}$	−0.5075	$\rightarrow \infty$	—
Local temperature rise estimate ΔT	−1.0100	$\rightarrow \infty$	(external estimate)

Table 3: Asymptotic scaling of key physical quantities for the OpenAI blow-up construction ($h = 1/200$). The global L^2 energy—the sole quantity mandated by the Millennium Prize—is the only quantity that remains bounded (last column: “yes”); every other row diverging is consistent with the Prize’s requirements, which impose nothing on local quantities (last column: “—”). Intensive local energy density $e_{\text{local}} = \frac{1}{2}\rho|u|^2 \sim \tau^{-1-2h}$ scales as $\tau^{-1.010}$, while the Sobolev $H^{3/2}$ norm scales as $\tau^{-0.5075}$. ΔT is an external estimate (§4), not a quantity solved for within incompressible Navier–Stokes.

The viscous dissipation rate is $\varepsilon(t) = \nu \int_{\mathbb{R}^3} |\nabla \times u|^2 dx \sim \tau^{-0.515}$. The Prize requires only that total dissipated energy remain finite, and indeed $\int_0^1 \tau^{-0.515} d\tau < \infty$ —so this constraint is satisfied. But this mathematical concession obscures the physical picture: the *instantaneous* dissipation rate ($\sim \tau^{-0.515}$) and the *intensive local energy density* ($e_{\text{local}} \sim \tau^{-1.010}$) both diverge without bound, and the *local* enstrophy density $|\omega|^2 \sim \tau^{-2-2h} = \tau^{-2.010}$ diverges much faster than its spatial integral. It is this divergence of local dissipation density—not the bounded integral—that controls the heating estimate of §5.

Incompressible Navier–Stokes carries no energy or temperature equation of its own: viscous dissipation is simply removed from the kinetic-energy budget, on the implicit assumption that the resulting heating is negligible (formally, that the relevant Eckert number stays small). That assumption is external to the model and has to be checked separately, not derived from it. Diverging local enstrophy means a diverging local viscous heating *rate*; §5 makes this quantitative for a specific fluid (water) and shows that the resulting heating estimate, together with the

velocity itself, invalidates the small-Eckert-number and incompressibility assumptions well before the mathematical singularity. This point was raised independently in community discussion of an early version of this audit (§11), where a commenter correctly noted that the thermal argument is an estimate built on top of the incompressible model from outside, not a consequence derivable within it — we agree, and flag deriving it properly (coupling an actual energy/temperature equation, or a cavitation/phase-change model) as an open direction in §11. A first step, a compressible Navier–Stokes–Fourier version of a forced collapsing core, is reported in §9.3.

The contrast with turbulence is worth stating without a figure. In physical turbulence the Kolmogorov cascade [22] ends at the microscale $\eta = (\nu^3/\varepsilon)^{1/4}$, which in every laboratory or geophysical flow exceeds the molecular length by orders of magnitude; the construction instead concentrates a vanishing fraction of the energy at a scale $\ell_r \rightarrow 0$ with no dissipative floor (see §8 on why no spectral figure is given here).

The L^3 norm also diverges, but only marginally ($\|u\|_{L^3} \sim \tau^{-4h/3} = \tau^{-0.0067}$ for $h = 1/200$): the Escauriaza-Seregin-Šverák criterion [7] proves a singularity *cannot* form while $\|u\|_{L^3}$ stays bounded, so this divergence — however small its exponent — is not a violation of anything; it is a necessary, and here barely satisfied, condition for the blow-up to be admissible at all. That the construction sits so close to the critical Serrin threshold is itself informative: it shows how little room the similarity ansatz leaves, and how sensitive the construction is to the precise value of h .

5 Where the Continuum Model Ends: One Scale, Four Limits

5.1 Unit-free core scales

The core of the construction keeps a radial Reynolds number of order one: with $\ell_r \asymp \tau^{1/2}$, $u \asymp \tau^{-1/2-h}$ and radial diffusion $\nu/\ell_r^2 \asymp \tau^{-1}$ in the leading balance [2, §2.1], the proof is written at viscosity one and τ is the dimensionless time-to-blow-up. Restoring units with an initial core radius ℓ_0 introduces the viscous time unit $T = \ell_0^2/\nu$ and the physical time-to-blow-up $t = T\tau$, and gives

$$\ell_r \simeq \sqrt{\nu t}, \quad u \simeq \sqrt{\nu/t} \left(\frac{\ell_0^2}{\nu t} \right)^h, \quad 0 < h < \frac{1}{100}, \quad (2)$$

with order-one prefactors that the paper does not make explicit. The initial size enters only through the factor $(\ell_0^2/\nu t)^h$, which for $h = 1/200$ lies between 1.2 and 1.3 over the whole range of interest and changes by a factor 1.3 when ℓ_0 changes by 10^6 . *The physical reading below is therefore essentially independent of how large the vortex was to begin with, and of the choice of units.* Earlier drafts of this audit reported numbers for $\ell_0 = 1$ cm ($T = 100$ s) and, at one point, read the dimensionless τ directly as seconds; Table 6 keeps the $\ell_0 = 1$ cm convention for continuity and reports $t = T\tau$.

5.2 Three validity limits are one

Proposition 5.1 (Scaling estimates, $O(1)$ prefactors). *Let c_s be the sound speed, c_p the specific heat, λ the molecular length (mean free path in a gas, intermolecular spacing in a liquid), and neglect the factor $(\ell_0^2/\nu t)^h$. Along the core:*

1. Compressibility. $\text{Ma} = u/c_s$ reaches a prescribed value at $t_{\text{Ma}} \simeq \nu/(\text{Ma}^2 c_s^2)$, when $\ell_r \simeq \nu/(\text{Ma} c_s)$.
2. Rarefaction. $\text{Kn} = \lambda/\ell_r$ reaches a prescribed value at $t_{\text{Kn}} \simeq \lambda^2/(\text{Kn}^2 \nu)$.
3. Viscous heating. The dissipation per unit mass is $\varepsilon \simeq \nu(u/\ell_r)^2$; since $\ell_r^2 \simeq \nu t$, its action over the remaining time t gives $\Delta T \simeq \varepsilon t/c_p \simeq u^2/c_p$, i.e. the Eckert number of the core is of order one.

For a gas, kinetic theory gives $\nu \simeq \frac{1}{2}\bar{c}\lambda$ with $\bar{c} \sim c_s$, so that $\text{Ma} \simeq \text{Kn} \cdot \text{Re}_r$ (von Kármán's relation); with $\text{Re}_r = O(1)$ all three effects become order one at the same length and time,

$$\ell_* \simeq \nu/c_s, \quad t_* \simeq \nu/c_s^2. \quad (3)$$

Equation (3) is the one-line physical content of this audit. In water, $\ell_* \approx 0.67$ nm (about two molecular diameters) and $t_* \approx 0.44$ ps; in air, $\ell_* \approx 45$ nm (essentially the mean free path) and $t_* \approx 0.13$ ns. The statement is generic: it uses only $\text{Re}_r = O(1)$, i.e. Leray-type self-similar scaling, and would apply to any Navier–Stokes singularity of this type, not only to OpenAI's construction. It does not apply to blow-up in general: scenarios in which the local Reynolds number grows or the velocity stays bounded meet other physics first (§9.4). Table 4 evaluates the conventional thresholds ($\text{Ma} = 0.3$, where density variations $\Delta\rho/\rho \approx \text{Ma}^2/2$ reach a few percent [25, 26]; $\text{Kn} = 0.1$, beyond which slip and non-continuum effects appear) for three fluids.

Fluid (300 K)	$\ell_* = \nu/c_s$	$t_* = \nu/c_s^2$	ΔT at $\text{Ma} = 0.3 / 1$	$ \omega _{\max} = c_s^2/\nu$
Water	0.67 nm	0.44 ps	48 K / 540 K	$2.2 \times 10^{12} \text{ s}^{-1}$
Air	45 nm	0.13 ns	11 K / 120 K	$7.5 \times 10^9 \text{ s}^{-1}$
Helium	121 nm	0.12 ns	18 K / 195 K	$8.3 \times 10^9 \text{ s}^{-1}$

Table 4: The one-scale content of Proposition 5.1: length ℓ_* , time t_* , viscous heating at the Mach limit, and the local admissibility bound of §5.7. Inputs at 300 K (ν , c_s , c_p): water $10^{-6} \text{ m}^2/\text{s}$, 1500 m/s, 4184 J/(kg K); air 1.56×10^{-5} , 343, 1005; helium 1.22×10^{-4} , 1007, 5193.

Fluid (300 K)	$t(\text{Ma} = 0.3)$	$t(\text{Kn} = 0.1)$	$t(\text{Kn} = 1)$
Water	5–7 ps	9 ps	90 fs
Air	1.5–1.8 ns	30 ns	0.3 ns
Helium	1.3–1.6 ns	27 ns	0.27 ns

Table 5: Times before blow-up at which each threshold is crossed. Ranges span the unit-free estimate $\nu/(\text{Ma } c_s)^2$ and the $\ell_0 = 1$ cm value of §5; molecular length λ : water 0.3 nm, air 68 nm, helium 180 nm.

5.3 The Mach trajectory for a 1 cm water vortex

For liquid water at reference ambient conditions ($T_0 = 300$ K, $c_s = 1500$ m/s, $\nu = 10^{-6} \text{ m}^2/\text{s}$), initialized with vortex radius $\ell_0 = 1$ cm and reference viscous velocity $u_0 = \nu/\ell_0 = 10^{-4}$ m/s, the collapse profile $u_\theta \sim u_0 \tau^{-1/2-h}$ with $h = 1/200$ yields the trajectory in Table 6; the constant-density idealization is used, by convention, while $\text{Ma} \lesssim 0.3$. The time column is $t = T\tau$ with $T = \ell_0^2/\nu = 100$ s.

5.4 Acoustic radiation is not an independent check

Lighthill's acoustic analogy [23, 24] gives, for a compact *subsonic* source of velocity U and size L , a radiated power $P_{\text{ac}} \sim \rho_0 U^8 L^2 / c_s^5$. Comparing the energy this would radiate over the remaining time t with the kinetic energy of the core, $E_{\text{core}} \sim \rho_0 U^2 \ell_r^2 \ell_z$, and using Eq. (2) for U , ℓ_r and $\ell_z \simeq \ell_r t^{-h}$, one finds

$$\frac{P_{\text{ac}} t}{E_{\text{core}}} \sim \left(\frac{U}{c_s} \right)^5 = \text{Ma}^5 \quad (4)$$

up to $O(1)$ constants and factors t^{-5h} . The time at which this ratio reaches unity is therefore not an independent check on the Mach limit: it is the Mach criterion again, expressed through the radiated power. The flow simply ceases to be an acoustically compact, subsonic source at $\text{Ma} \sim 1$ — exactly where Eq. (3) says the incompressible model has ended.

Time $t = T\tau$ (s)	Velocity $ u $ (m/s)	Mach Ma	Core Radius ℓ_r (m)	Physical Regime
1.00×10^2	1.00×10^{-4}	6.67×10^{-8}	1.00×10^{-2}	Incompressible continuum
1.00×10^{-4}	1.07×10^{-1}	7.14×10^{-5}	1.00×10^{-5}	Incompressible continuum
1.00×10^{-10}	1.15×10^2	7.65×10^{-2}	1.00×10^{-8}	Laminar high-speed core
6.69×10^{-12}	4.50×10^2	0.30	2.59×10^{-9}	Incompressibility limit breached
6.16×10^{-13}	1.50×10^3	1.00	7.85×10^{-10}	Sonic barrier
9.00×10^{-14}	3.96×10^3	2.64	3.00×10^{-10}	Mean free path limit (Kn = 1)
1.00×10^{-16}	1.23×10^5	82.0	1.00×10^{-11}	Sub-molecular regime
1.00×10^{-23}	4.22×10^8	2.81×10^5	3.16×10^{-15}	Formally superluminal — no longer meaningful

Table 6: Trajectory of the OpenAI collapsing vortex core in water ($c_s = 1500$ m/s, $T_0 = 300$ K), with the time column corrected to physical seconds $t = T\tau$, $T = \ell_0^2/\nu = 100$ s. The incompressible model self-invalidates at $t \approx 6.69 \times 10^{-12}$ s (≈ 6.7 picoseconds before the mathematical singularity), and crosses the sonic barrier at $t \approx 6.16 \times 10^{-13}$ s. Velocity, Mach and radius columns are functions of the dimensionless τ only and are unaffected by the time-unit correction. The last two rows extrapolate the construction’s own scaling law past the point (row 4 onward) where it has any physical referent; they illustrate the mathematics of the construction, not a physical trajectory.

5.5 Viscous heating: the temperature the model cannot see

Incompressible Navier–Stokes carries no temperature equation (§4), so the heating is an external estimate; Proposition 5.1(3) makes it a sharp one. Because the core radius is the viscous length $\sqrt{\nu t}$, the heat released by viscous dissipation over the remaining time equals the local kinetic energy: $\Delta T \simeq u^2/c_p$. For water this is 48 K at Ma = 0.3 and 540 K at Ma = 1; the boiling point ($\Delta T = 73$ K) is reached at Ma ≈ 0.37 , about 3–4.5 ps before blow-up. Earlier drafts of this audit spoke of “plasma temperatures”; a rise of 10^4 K would require Ma ≈ 4 , far past the point where the model has ended, and the $\tau^{-1.010}$ law of Table 3 must not be extrapolated there. Two caveats: the estimate assumes the heat stays in the core over the collapse time, which holds when the thermal diffusion length $\sqrt{\alpha t} = \text{Pr}^{-1/2} \ell_r$ does not exceed ℓ_r — true for water ($\text{Pr} \approx 7$), marginal for air ($\text{Pr} \approx 0.7$), where ΔT is an upper estimate (a compressible simulation with an energy equation finds a coefficient of 0.35 at Ma ≈ 0.4 , rising to about one near Ma ≈ 1 ; §9.3); and the model’s own energy balance is exact and its dissipation is positive, so *no thermodynamic law is violated* — what fails is the constitutive assumption of a decoupled, constant temperature.

5.6 Liquids: cavitation comes first

For a liquid at ambient pressure the first constitutive assumption to fail is neither incompressibility nor the continuum but the absence of a vapour phase. The pressure at the centre of a vortex core lies below the ambient pressure by roughly ρu^2 (Bernoulli), so the local pressure reaches the vapour pressure when $\frac{1}{2}\rho u^2 \approx p_\infty - p_v \approx 0.1$ MPa, i.e. at $u \approx 14$ m/s (Ma ≈ 0.01). By Eq. (2) this happens $t \simeq \nu/u^2 \approx 5$ ns before blow-up, when the core radius is ≈ 70 nm — three orders of magnitude earlier in time than the Mach limit, in a core still a hundred molecules wide. A nanometre-scale core that lives for picoseconds may not nucleate at p_v : the measured tensile strength of clean water is ~ 30 MPa, and ~ 140 MPa has been reached in mineral inclusions [43, 44]. Those tensions correspond to $u \approx 245$ – 530 m/s, $t \approx 17$ – 4 ps and $\ell_r \approx 4$ – 2 nm — before or together with Ma = 0.3. Whether the liquid cavitates at 5 ns or holds tension to 5 ps, it stops being the fluid the incompressible model describes before, or at, the limits of

Proposition 5.1. For gases there is no such limit and Eq. (3) stands alone.

These figures take the core pressure deficit to be $\frac{1}{2}\rho u^2$. For a Gaussian (Lamb–Oseen) core the deficit is exactly $(\ln 2/0.638^2) \rho u_{\max}^2 = 1.70 \rho u_{\max}^2$, which brings the threshold forward to $u \approx 7.6$ m/s, a core of about 130 nm and 17 ns before blow-up; the construction’s profile is not Gaussian, so the two values bracket rather than fix the number. Once a vapour core opens, a hollow vortex caps the swirl speed at $\sqrt{2(p_\infty - p_v)/\rho} \approx 14$ m/s: for a liquid, the first thermodynamic event is a phase change, and it bounds the velocity.

5.7 A single admissibility condition

The limits of Proposition 5.1 combine into one local condition. A vortex of velocity U and size L has $|\omega| \sim U/L$; validity requires $U \lesssim c_s$ (Mach) and $L \gtrsim \nu/c_s$ (Knudsen, through ℓ_*), hence

$$|\omega| \lesssim \frac{c_s^2}{\nu}, \quad \text{equivalently} \quad \varepsilon = \nu|\omega|^2 \lesssim \frac{c_s^4}{\nu}, \quad (5)$$

about $2.2 \times 10^{12} \text{ s}^{-1}$ ($\varepsilon \lesssim 5 \times 10^{18} \text{ W/kg}$) in water and $7.5 \times 10^9 \text{ s}^{-1}$ in air. This local, dimensionally derived, material-specific bound is deliberately distinct from a *global* integral enstrophy bound, which scales with the volume of fluid considered and is not dimensionally forced the way Eq. (5) is (see `05_Community_Research_Directions/WorkStream1_EntropyCondition` for the companion admissibility discussion). It also turns the admissibility statement from a definition into a theorem. By Beale–Kato–Majda [8], which holds for Navier–Stokes as well as Euler, a solution with $\int_0^T \|\omega(t)\|_{L^\infty} dt < \infty$ is smooth on $[0, T]$; a solution that satisfies Eq. (5) on $[0, T)$ therefore does not blow up at T , and any blow-up must first leave the admissible set — in this construction at ℓ_* , or in a liquid at the cavitation scale. The condition says nothing about whether real flows approach the bound; it says where the model must be replaced by another. In local variables it reads $\text{Ma}^2 \leq \text{Re}$ (§9.4), and that OpenAI’s candidate objects necessarily violate it near $t = 1$ is proved in Lean on their own definitions (Direction 2, §11.5).

5.8 What laboratory vortices do instead

Laboratory experiments on interacting vortex filaments in water [27] show reconnection rather than self-similar collapse: core lines approach, pinch off and rearrange, shedding energy into waves and small-scale motion. That is a different configuration from the single swirling column of the construction, and it does not contradict the theorem; it is a reminder that real vortices find other routes to dissipation long before molecular scales. Likewise, extreme dissipation events in direct numerical simulations and high-Reynolds-number experiments [28, 29] are intense but finite and resolved at the Kolmogorov scale η , which in every measured flow exceeds the molecular length by orders of magnitude. The Beale–Kato–Majda criterion [8] is the mathematical statement that this finiteness is what prevents blow-up; it is not itself a statement about molecules.

6 Backward-Specified Forcing and the Limits of MMS

Physics problems are usually posed forward: specify a force, then compute how the fluid responds. This proof’s logical structure runs the specification backward. The authors:

1. **Stipulate** the goal: a singularity at $t = 1$.
2. **Engineer** a collapsing vortex column whose velocity diverges (Section 4 of [2]).
3. **Compute** the raw residual, which is singular ($f_{\text{raw}} \rightarrow \infty$).
4. **Construct** oscillatory wave packets (Sections 6–9) whose internal Reynolds stresses cancel the singular residual.

5. **Declare** the remaining smooth remainder as the external force $f(x, t)$ (Section 10).

This is a completely legitimate way to build an existence proof — fixing the endpoint and solving backward for compatible data is standard technique — but it has a specific consequence worth flagging: because (u, f) are co-designed as a single package, f cannot be given an independent physical origin (a stirring mechanism, a boundary forcing, a body force from an external field) without already knowing the singular solution it is meant to produce. The paper is transparent about this itself: “An exponentially small external force seeds each pulse; the background shear supplies its subsequent growth.” The force *seeds* — it does not *drive*.

It is, in that sense, a demonstration that the singular endpoint *can* be reached by a smooth force, not a claim about which forces occur in nature or where flows normally go.

7 The Unforced Euler Singularity: Coherent Structure Below Any Physical Cutoff

For the unforced Euler equations the singularity must come from the initial datum alone. The construction [4] builds $u_0 \in C_{c,\sigma}^\infty(\mathbb{R}^3)$ by a stagewise iteration: at each stage a localized oscillatory packet of higher frequency is added to the flow built so far, the packet is amplified by the shear of that parent flow, and its averaged momentum flux feeds the next stage; amplitudes, frequencies, localization lengths and time intervals are chosen so that the sum converges in every fixed Sobolev norm while $\int_0^{T_*} \|\omega(t)\|_{L^\infty} dt$ diverges at a finite T_* ([4, §§2–5]).

Because u_0 is C^∞ , its energy below any molecular length λ is negligible; the physical issue is not energy but *coherence*. The mechanism requires the packets, down to arbitrarily fine scales, to be present in the datum with prescribed phases so that the shear of the coarser stages can amplify them in the right order. In a real fluid the Euler equations are the $\nu \rightarrow 0$ idealization of Navier–Stokes, so the physics missing from the Euler construction is viscosity itself, which damps a packet of wavenumber k on the time $1/(\nu k^2)$ — together with thermal fluctuations, which set an irreducible noise floor near the scale $\ell_{\text{th}} = k_B T / (\rho \nu^2)$ (a few picometres in water, i.e. already below λ , so the molecular length is the operative cutoff [41]). Whether a viscous, noisy fluid can reproduce the first few stages of the cascade before the fine scales are washed out is a computable question (§11), not a slogan. Earlier drafts of this audit described the fine scales as “sub-Planckian”; the Planck length is irrelevant here — the cutoff is molecular.

A recent human-written construction makes the point sharper. For the Euler equations with a force that is smooth up to and including the blow-up time, [10] obtains a singularity in which the circulation and the meridional velocity remain bounded while the gradient of the circulation and the vorticity diverge. In a blow-up of that kind the Mach number never grows, and the scale at which a real fluid departs from it is the viscous one, $\nu/u = \ell_*/\text{Ma}$, which for a subsonic flow lies *above* ℓ_* (§9.4).

8 Properties of the Specific Construction, and What Was Removed

Table 7 collects properties of the particular solution built. They are properties of that solution, not defects in the proof.

8.1 What this paper does not contain, and why

No numerical realization of the 166-page OpenAI construction exists, in this project or, to our knowledge, elsewhere. The scripts behind §§4–7 are exponent, unit and consistency checks on the scaling laws above, and the simulations of §9 are of a surrogate — a manufactured Gaussian core with the construction’s diffusive scaling — and are labelled as such; none is a

Property	OpenAI formalization	Reading
1. Backward-specified residual	<code>force</code> defined via <code>SpacetimeGluing.smoothExtension</code> of <code>tracedResidual</code> (<code>CandidateFromLimits.lean</code>)	The force is the smooth remainder of a co-designed (u, p) (§6); initial data is $u_0 \equiv 0$ for the forced case.
2. One explicit datum	$\exists u_0, \text{ContDiff } \mathbb{R} \propto u_0$	A single point of $C_{c,\sigma}^\infty$. Its codimension — the number of unstable directions of the linearized self-similar profile beyond the symmetry modes — is the rigorous measure of “non-generic” and is not established by this audit (§11).
3. Axisymmetric swirl	<code>AxisymmetricFields.lean</code>	A 2D similarity profile with swirl; it need not represent generic three-dimensional turbulence.
4. No dissipative floor	Corrections summed with shrinking cutoffs down to $\ell_r \rightarrow 0$	There is no Kolmogorov scale η . A solution that stayed within Eq. (5) could not blow up (Beale–Kato–Majda); whether any physical mechanism enforces the bound is a separate question, tested in §9.
5. Engineered lifetime	<code>SmoothCutoffs.lean</code> time switch	Forcing is compactly supported in time; the lifespan $T^* = 1$ is a choice, not a feature of the equations.

Table 7: Properties of the constructed solution, with the Lean locations that encode them.

simulation of the construction itself. Two figures that appeared in earlier public drafts of this audit are deliberately absent here: a Taylor–Green-vortex comparison whose resolution and normalization were not documented to a reproducible standard, and a spectrum plot contrasting real turbulence data with a hand-drawn schematic of an “OpenAI pre-singularity spectrum” rather than a computed one. Neither met the standard we hold the rest of this paper to, so neither is reproduced. Appendix A records this and the other differences from earlier versions in one place. What a genuine numerical companion should contain beyond §9 is listed in §11.5: the construction’s own profile, the codimension count and the fluctuating-hydrodynamics questions.

9 Testing the Cutoff Hypothesis: What Acts Where the Model Ends

We implemented a three-dimensional pseudo-spectral Navier–Stokes solver (rotational form, FFT Leray projection, Orszag 2/3 dealiasing, integrating-factor RK4 so that the stiff barrier term imposes no step restriction) and swept the regularization parameter. Code, data and reproduction instructions accompany this paper. The solver reproduces the Taylor–Green transition benchmark’s dissipation peak at $t = 9.14$ against a published $t = 9.0$ at $\text{Re} = 1600$, with the peak magnitude 7% high in the manner expected of an under-resolved spectral run ($k_{\max}\eta = 0.25$).

Three things came out of it, and the third is the useful one.

The law is exact, given its premise. Granting a core on the diffusive scaling $\ell_r = \sqrt{\nu t}$, $u = \sqrt{\nu/t}$ — equivalently $\text{Re}_{\text{core}} = u\ell_r/\nu = 1$ — the arrest condition follows from the barrier operator itself rather than by assumption: the operator $\nu k^2 \max(1, \alpha' k^2)$ exceeds the plain viscous rate at the core’s own wavenumber exactly when $\ell_r < \sqrt{\alpha'}$. Substituting back reproduces

all four exponents as identities, to machine precision. A disagreement with experiment is therefore a statement about the premise, not about the arithmetic.

The premise is not reached. In a forced dyadic cascade swept over 3.6 decades in α' , the measured exponents are $|\omega|_{\max} \sim \alpha'^{-0.192}$ (predicted -1) and $u \sim \alpha'^{+0.173}$ (predicted $-1/2$, and the sign is wrong), with $R^2 = 0.98\text{--}0.99$. Written in the barrier wavenumber $k_\alpha = \alpha'^{-1/2}$, the measured velocity at the arrest scale is $u \sim k_\alpha^{-0.345}$, which matches the Kolmogorov inertial-range value $k^{-1/3}$ to 3.5% and misses the cutoff law's $u \sim \nu k_\alpha$ by three orders of magnitude. The local Reynolds number at the arrest scale runs from 1.2×10^3 down to 4.9 across the sweep, never approaching unity. The cutoff law describes a viscously dominated $\text{Re} \approx 1$ core; a cascade places the arrest scale in an inertial range, where inertial-range scaling governs and the law has no purchase.

Direct simulation cannot close the gap either. The same test in three dimensions is bounded from both sides. The barrier caps vorticity at $\nu/\alpha' = \nu k_\alpha^2$, so arresting a *growing* vorticity rather than merely damping the initial field requires $\nu k_\alpha^2 > |\omega|_0$, while resolution requires $k_\alpha < k_{\max} = n/3$ and the barrier must engage before ordinary viscosity does, $k_\alpha < k_\eta$. The first two give

$$n > 3\sqrt{|\omega|_0 \text{Re}}, \quad (6)$$

that is $n > 85$ at $\text{Re} = 400$ and $n > 170$ at $\text{Re} = 1600$ for Taylor–Green data. This is why sweeps at 64^3 in earlier work on this project moved the peak enstrophy by less than a tenth of a percent across five decades of α' : the barrier was never engaged at all. At 96^3 and $\text{Re} = 200$ the admissible band is empty outright — its lower edge $\sqrt{|\omega|_0/\nu} = 20$ exceeds the measured $k_\eta = 17.6$ — and every run in the sweep excluded itself, reporting which control it failed.

The sharpened question. The cutoff law cannot be tested with generic initial data at any resolution, because generic initial data does not produce a $\text{Re} \approx 1$ collapsing core; that core is a specific property of the construction, sustained by its forcing. A genuine test therefore requires reproducing the construction's own forcing, or a truncation of it, rather than any flow that merely develops small scales. We regard this as a more useful statement of the open problem than the one it replaces, and note the corollary: for $\sqrt{\alpha'} \gg \eta$ the inertial range governs, while at $\sqrt{\alpha'} \sim \eta$ the barrier acts where viscosity already does and α' stops being an independent parameter. The window in which a cutoff regularization could say anything new about a physical fluid is correspondingly narrow.

A distinct regularization: Leray- α gates transfer rather than draining energy. The regularization tested above modifies *dissipation*, $\nu|k|^2 \mapsto \nu|k|^2 \max(1, \alpha'|k|^2)$. The Leray- α model [35, 40] modifies *transport* instead,

$$\partial_t u + (\bar{u} \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \bar{u} = (1 - \alpha^2 \Delta)^{-1}u, \quad (7)$$

and it is this second family, not the first, whose three-dimensional global well-posedness is a theorem. The two are easily conflated, and earlier documents in this programme did conflate them; the negative result above concerns the first. It transfers to the second only in its kinematic part: arrest at the filter scale for a $\text{Re} \approx 1$ core still gives $t_c \sim \alpha^2/\nu$ and $u_{\max} \sim \nu/\alpha$, and generic data still does not supply that core.

What does not transfer is the mechanism, and here the two regularizations differ in kind. The Leray- α nonlinearity does no work on u : since $\bar{u}$ is solenoidal, $\int u \cdot (\bar{u} \cdot \nabla)u \, dx = -\int (\nabla \cdot \bar{u})|u|^2/2 \, dx = 0$, so the inviscid system conserves $\frac{1}{2} \int |u|^2$ exactly (to a drift of 3×10^{-16} over two hundred steps in our implementation). Implementing Eq. (7) in the same solver, on the same Taylor–Green data at $\text{Re} = 200$, peak vorticity falls monotonically with α , by a factor of

3.3 between $\alpha = 0$ and $\alpha = 0.4$, while the fraction of energy dissipated *decreases* from 55% to 35%; the hyperviscous barrier, by contrast, dissipates slightly more than plain Navier–Stokes. Hyperviscosity destroys energy that reaches small scales; Leray- α prevents it from arriving. One is a drain and the other a gate. Neither result bears on the regularity of the Navier–Stokes equations themselves — a theorem about a modified equation does not constrain the unmodified one. Which of the two acts more strongly turns out to depend on the Reynolds number of the flow they act on, and §9.2 measures that dependence on the collapse the cutoff law is actually about; whether either represents what a real fluid does at ℓ_* is a different question, answered in the negative in §9.2 and §9.6.

9.1 Supplying the premise: a forced collapsing core

The obstruction above is the absence of a $\text{Re} \approx 1$ collapsing core in generic flow, so we manufacture one. The target is a time-reversed Lamb–Oseen column: Gaussian vorticity $\omega_z = \Gamma/(\pi\ell^2) e^{-r^2/\ell^2}$ with fixed circulation and $\ell^2 = \nu\tau$, the diffusive scaling of Eq. (2). Choosing $\Gamma = 2\pi\nu/0.638$ gives $u_{\max}\ell/\nu = 1$ exactly, so the cutoff law’s premise holds by construction. The forcing is the residual of the target under the plain Navier–Stokes operators, $f = \partial_t U - N(U) - \nu\Delta U$; for the column $N(U)$ is a pure gradient and projects out, and $f = -\frac{5}{4}\nu\Delta U$ — time-reversed diffusion requires anti-diffusion. With the regularization off, the simulated flow tracks the analytic target to a whole-field relative error of 1.7×10^{-7} over the entire collapse.

With the barrier on and the *same* forcing, the natural diagnostic is the ratio B/F of the barrier’s extra dissipation to the forcing’s power input. Scaling gives $B/F \propto \alpha' / (\nu\tau)$, and arrest is the crossing $B/F = 1$. Over a $2.6\times$ range of α' the measured B/F *collapses onto a single function of $\alpha' / (\nu\tau)$* : the per-run prefactor is 0.188 ± 0.010 , and at a common value of the variable the four runs agree to 6%. That collapse is the load-bearing result, because it is sufficient on its own: if B/F depends on $\alpha' / (\nu\tau)$ alone, arrest occurs at a fixed value x_* of it, so $\tau_c = \alpha' / (\nu x_*)$ and all four predicted exponents follow with no assumption about the curve’s shape. The curve is not a pure power law — its pooled log-log slope is 0.85, falling from 1.13 to 0.56 across the band — and we checked the obvious alternative explanation before accepting this: a lagging core would steepen the curve, and refitting against the measured rather than the target core size gives a slope of 2.4, the wrong direction. The shape is instead that of the fraction of the forcing’s Gaussian spectrum lying above k_α . At 32^3 the crossing itself falls just below the resolved window. We expected a 96^3 sweep to reach it; it did not — none of six runs attains $B/F = 1$, and B/F saturates at 0.25–0.56. The reason is informative: the barrier resists by fattening the core, which lowers its own share of the budget, so the ratio levels off instead of crossing. The arrest is visible in the core size instead. With the barrier off the core follows $\ell \propto \tau^{1/2}$ exactly (local slope 0.500); with it on, every run decelerates to a local slope of 0.04–0.16 at $\ell \approx (1.2\text{--}1.5)\sqrt{\alpha'}$ by the resolution floor, and the window-end exponents against α' are +0.42 for ℓ and -0.43 to -0.47 for $u_{\max}$, approaching the law’s $+1/2$ and $-1/2$ from below. No run has fully stalled, so this supports $\ell_c \propto \sqrt{\alpha'}$ without yet establishing a converged arrest.

9.2 Gate or drain on a $\text{Re} \approx 1$ collapse

On the swirl column the α -model nonlinearities are also pure gradients, so Leray- α and LANS- α [36] are inert there. Adding an axisymmetric meridional cell with axial Reynolds number one makes them act, and the same forcing construction then tracks its target to 6.5×10^{-7} . On this target, with filter width matched to the barrier scale ($\sqrt{\alpha'} = \alpha = 0.25\text{--}0.5$), the transport filters lag the collapse by 0.2–0.4% with no change to its rate, while the barrier lags it by 5–45%, roughly halves $d\ell/dt$, and cuts the swirl-to-axial energy transfer threefold.

The reason is visible in the size of each term measured against the forcing. Viscosity is about 0.85 of it; the projected nonlinearity is 0.04–0.05. A $\text{Re} \approx 1$ collapse is a balance between forcing and viscosity, and a transport filter can act only through the nonlinear share, so its leverage

is capped by a quantity that is small *because* the Reynolds number is one. The barrier acts on the field directly. Scaling the target by λ multiplies the core Reynolds number by λ while linear terms scale as λ and the nonlinearity as λ^2 , which gives an exact scan without further simulation (Table 8).

core Re	nonlinear / forcing	transport filter / forcing	barrier / forcing
0.3	0.014	0.006	0.83
1	0.046	0.020	0.83
10	0.42	0.18	0.76
30	0.81	0.35	0.49
100	0.98	0.43	0.18
1000	1.00	0.44	0.018

Table 8: Leverage of each regularization, as the norm of the perturbation it makes to the dynamics divided by the norm of the forcing, for a Leray- α filter and a barrier at the same scale ($\alpha = 0.35$, $t = 0.4T$). LANS- α agrees with Leray- α to within 5% throughout.

The Reynolds number at which the two are equal needs no fitting. Both leverages are divided by the same forcing norm, which cancels in their ratio, so the crossover is exactly

$$\text{Re}_\times = \frac{\|(L_{\text{barrier}} - L_\nu)U\|}{\|N_\alpha(U) - N(U)\|}, \quad (8)$$

a property of the target and the filter alone. It is not universal: over three filter widths and three core sizes it ranges from 8 to 142, rising steeply as the filter width approaches the core size. It is well above one in every case, and largest as $\alpha/\ell \rightarrow 1$, which is where arrest happens.

Which lock acts where the continuum ends. Proposition 5.1 places a core with $\text{Re}_r = O(1)$ at ℓ_* , and arrest there means $\alpha/\ell \rightarrow 1$ — precisely the regime in which a transport filter has two orders of magnitude less leverage than dissipation. We had conjectured that LANS- α with $\alpha \approx \ell_*$, being a Lagrangian average, might be *derived* rather than posited as the regularization at the continuum limit. This measurement first relocated that conjecture: a transport filter is the stronger regularization for transfer-driven dynamics at $\text{Re} \gtrsim 10$ –100, the regime where the cutoff law’s premise fails, and the weaker one on the collapse. The identification of the filter width with ℓ_* is withdrawn altogether in §9.6. (Both regularizations are models; the comparison ranks them against each other, and says nothing yet about which, if either, resembles what a real fluid does at ℓ_* . That question is taken up below, and the answer is neither.) It is consistent with Tao’s averaged Navier–Stokes blow-up [18], which preserves the energy identity and shows that what prevents blow-up must constrain the structure of transfer, not the energy budget — but on a $\text{Re} \approx 1$ core there is little transfer to constrain.

What does act there is set by the physics beneath the continuum, and the first step is to identify that physics precisely. Kinetic theory gives $\nu \approx \frac{1}{2}\bar{c}\lambda$ for a gas, so $\ell_* = \nu/c_s \approx (\bar{c}/2c_s)\lambda$; for air $\bar{c}/2c_s = 0.67$, and the independently computed $\ell_* = 45$ nm against a mean free path of 68 nm gives the same 0.67. The validity scale of Proposition 5.1 *is* the mean free path, with a derived constant. The hierarchy connecting the two descriptions is Chapman–Enskog, with the Knudsen number as the locking parameter, and incompressible Navier–Stokes is a rigorous limit of the Boltzmann equation [37]. The collision operator that takes over below λ is dissipative in the thermodynamic sense: for its discrete BGK form [39] we have proved in Lean that each collision step contracts the relative entropy by at least a factor $1 - \omega$ (§11.3).

It is tempting to conclude that the kinetic regime acts like the barrier — a stronger drain on small scales. It does not, and the exact linear theory shows why. For the BGK model with collision time τ (so $\nu = v_{\text{th}}^2\tau$ and $\lambda = v_{\text{th}}\tau$), a transverse shear perturbation of wavenumber k

obeys the dispersion relation

$$1 = \frac{1}{\tau} \left\langle \frac{1}{\sigma + ikv} \right\rangle_{v \sim \mathcal{N}(0, v_{\text{th}}^2)}, \quad \sigma = s + 1/\tau, \quad (9)$$

which we solved exactly through the Faddeeva function and, independently, as the spectrum of a discrete-velocity operator; the two agree to 3×10^{-12} . Three facts follow. At small k the shear damping rate is $\Gamma = \nu k^2 [1 - (k\lambda)^2 + \dots]$: the first kinetic correction *reduces* the damping, the opposite sign to hyperviscosity, and truncated at that order it turns negative for $k\lambda > 1$ — the Burnett instability [38], which the full kinetic mode does not share. The damping rate never exceeds the collision rate $1/\tau$; this is exact, since the linearized operator is a skew advection term plus $-(1/\tau)(I - P)$ with P a projection, the spectral counterpart of the H-theorem. And at

$$k\lambda = \sqrt{\pi/2} \approx 1.2533 \quad (10)$$

the hydrodynamic shear mode ceases to exist: as $\sigma \rightarrow 0^+$ the velocity average in Eq. (9) tends to $\sqrt{\pi/2}/(k\lambda)$, so no root survives beyond it and only free-streaming decay of the continuum remains. At $k\lambda = 1$ the barrier already damps 1.43 times harder than kinetic theory does; at $k\lambda = 3$ there is no kinetic mode to compare, and the barrier exceeds the collision-rate cap by a factor of 81 or more.

That is a statement about linear modes. Whether a *driven* nonlinear core is stopped near $k\lambda \approx \sqrt{\pi/2}$ is a separate question, and we tested it directly with a two-dimensional, two-velocity discrete-velocity BGK solver (entropic discrete Maxwellian, exact collision, spectral streaming, Strang splitting), forced by the same manufactured collapse. Before any collapse run it reproduces the equilibrium moments to 10^{-15} , conserves mass and momentum to 2×10^{-15} with the discrete H decreasing at every step, agrees with an independent CVODE integration at second order, keeps $f \geq 0$ in Taylor–Green flows up to Mach 0.6, and matches the exact shear decay rates of Eq. (9) within 2% for $k\lambda \leq 1$. The kinetic core does not arrest. It runs slightly *ahead* of the Navier–Stokes target (by up to 12% in core size), as a model with weaker small-scale damping should, and its apparent stopping point moves with the grid rather than with λ : at fixed λ , refining the grid spacing from 0.67λ to 0.17λ moves it from 0.98λ to 0.52λ at $\text{Re} = 1$ and from 0.96λ to 0.35λ at $\text{Re} = 0.25$, with $k_\omega\lambda$ at that point rising to 4–5. A Navier–Stokes control on the same grid and forcing tracks its target to 4×10^{-4} . Within the solver’s validated range the robust statement is that there is no arrest at or above $\ell \approx 0.9\lambda$; below it the runs reach Mach numbers of 1–3, strong density depletion and small negative populations, and bound rather than measure what happens. The model is isothermal, so it contains no thermodynamic limit by construction. This is consistent with the linear result rather than in tension with it: a model whose damping is capped at $1/\tau$ offers a body force no additional resistance to overcome.

So what acts at ℓ_* on this route is neither of the two regularizations tested above. It is not a stronger drain, and not a gate on transfer. It is the end of the hydrodynamic description: the velocity gradient on which Navier–Stokes, the barrier and the α -models all act stops having autonomous dynamics at a scale fixed by the mean free path — a termination, not an arrest. That returns this paper to the reading of §5, now with a mechanism rather than only a scale. A phenomenological Mach cap applied to a dyadic cascade makes a complementary point, and anticipates §9.4: in a cascade it engages at the *largest* scale, while on the diffusive collapse it engages at exactly ℓ_* .

9.3 Compressibility and thermodynamics on the forced core

The kinetic test above excluded more than it appeared to. A BGK gas with constant collision time has $\mu \propto \rho$, hence constant ν ; a real gas has $\tau \propto 1/\rho$, so μ is independent of density and $\nu = \mu/\rho$ rises as the core evacuates. That feedback, and the energy equation, were both absent by construction. We therefore applied the same force — the residual of the *incompressible, constant-viscosity* equations, which is also the model OpenAI’s force is built for — to an axisymmetric

compressible Navier–Stokes–Fourier fluid, one-dimensional in radius, switching on one piece of physics at a time: the viscosity law ($\mu \propto \rho$, μ constant, $\mu \propto T^{0.76}$), the thermodynamics (isothermal, or the full energy equation for air with $\gamma = 1.4$, $\text{Pr} = 0.71$ and viscous heating), and the force convention (per unit mass or per unit volume). The low-Mach control tracks its target to 7×10^{-6} , and all numbers below are unchanged to three digits on doubling the grid.

For the Gaussian core the cyclostrophic density hole is exact, $\ln(\rho_0/\rho_\infty) = -(\ln 2/0.638^2) \text{Ma}^2 = -1.70 \text{Ma}^2$, and a quasi-steady argument then predicts arrest at $\text{Ma} = 0.36$, where $\nu/\hat{\rho}$ overtakes the force’s anti-diffusion $(5/4)\nu$. *The prediction fails*: no run arrests. At $\text{Re} \approx 1$ the hole cannot form fast enough — evacuating it quasi-steadily needs a radial outflow $u_r/c_s \approx 1.7 \text{Ma}^3/\text{Re}$, supersonic by $\text{Ma} \approx 0.8$ — so the collapse outruns its own compressible response. With the target driven to $\text{Ma} = 2$ at $\text{Re} = 1$, the core lags by 14% with the real-gas viscosity law and by 25% with the full thermodynamics of air (core temperature $1.44 T_\infty$), and runs 5% *ahead* with constant ν , the continuum twin of the kinetic run. Heat acts on the collapse only through the viscosity law: with the energy equation on but μ held fixed the lag returns to 14%. On the diffusive route, then, neither compressibility nor heat stops the driven core before ℓ_* ; they slow it, and the state they produce has a local Knudsen number of 4–8, outside the model that produced it. (The heating estimate $\Delta T \simeq u^2/c_p$ of §5 is an upper-order estimate: the measured coefficient is 0.35 at $\text{Ma} \approx 0.4$, rising to about one near $\text{Ma} \approx 1$.)

model (force per unit mass unless stated)	core Reynolds number				
	0.25	1	4	16	64
ν constant, isothermal	3.88	3.16	2.11	1.91	1.90
real-gas μ , isothermal, per volume	4.16	4.66	4.95	4.98	4.98
real-gas μ , isothermal	3.37	1.97	1.20	1.10	1.11
air, full thermodynamics	1.11	0.92	0.74	0.70	0.70

Table 9: Peak *local* Mach number of the forced core at the instant the incompressible target reaches $\text{Ma} = 4$. Driving the target on to $\text{Ma} = 8$ at $\text{Re} = 16$ leaves air at 0.70 and moves the isothermal real gas from 1.10 to 1.22. With the energy equation the lock also holds for a force per unit volume (0.81 at $\text{Re} = 16$), where the isothermal case runs away and evacuates its core to $\rho_0 = 10^{-4}\rho_\infty$.

Along the inertial route the picture changes (Table 9). For $\text{Re} \gtrsim 16$ — where the compressible continuum model is valid, the local Knudsen number being 0.25 at $\text{Re} = 64$ — a gas with an energy equation does not follow the driven swirl past a local Mach number of 0.70, however far the target is driven. The work of the force goes into heat and into evacuating the core: at $\text{Re} = 16$, as the target goes from $\text{Ma} = 1$ to 8, the core temperature rises from 1.10 to $2.03 T_\infty$, its density falls from 0.38 to $0.12 \rho_\infty$, its kinematic viscosity rises fifteen-fold, and the swirl Mach number stops rising. This is the first positive “lock” result of this programme, and three qualifications belong with it. It is a lock on the *Mach number*: the velocity still creeps up as $0.70 c_s(T_0)$, and the core still shrinks, at half the unimpeded rate. It is a statement about an open-loop force built for the incompressible model, in a one-dimensional axisymmetric ideal gas with constant γ . And it is not a regularity statement: the compressible Navier–Stokes equations have their own proved finite-time singularities, by implosion [11, 12], so each model in the hierarchy hands its breakdown to the next. A cross-check in the kinetic solver with $\tau \propto 1/\rho$ agrees in direction but is small where that solver is valid (peak Mach 0.72 against 0.77 for constant τ at $\text{Re} = 3$, target $\text{Ma} = 0.75$); the solver is isothermal and limited to $\text{Re} \leq 3$ and to Mach numbers below about 0.6.

A closure-free test by molecular dynamics. The lock was therefore tested with no continuum closure at all: molecular dynamics of a Lennard-Jones gas ($\rho = 0.15 \sigma^{-3}$, $T = 2 \epsilon/k_B$, mean free path 1.49σ ; 107,584 particles in a $300 \times 300 \times 8 \sigma$ slab), driven by the same per-

unit-mass force, with a thermostat acting only beyond 3.2 initial core radii. No viscosity law, heat-conduction law or equation of state is put in. The simulation reproduces the gas it models — energy drift below 4×10^{-5} , pressure within 0.8% of the third-virial equation of state, and a viscosity measured in situ as 1.71 ± 0.07 (decaying shear waves) and 1.87 ± 0.18 (a decaying vortex) against the modified-Enskog value 1.83. The continuum prediction for this set-up was registered before the molecular data existed (local Mach number saturating near 0.58 at $\text{Re} = 16$). Local Mach numbers in the molecular gas are formed with the sound speed $c(\rho, T)$ of its own equation of state, bin by bin, which matters: the ideal-gas value $\sqrt{\gamma T}$ is 8.5% low at ambient density though only 1–2% low in an evacuated core. At $\text{Re} = 16$ (three independent runs, target driven from Mach 0.34 to 3.4) the two descriptions agree where the gas is still a continuum: at target Mach 1 (local Knudsen number ≈ 0.2) the peak local Mach number of the molecular core is 0.538 ± 0.011 from a Lamb–Oseen fit (0.61 from the maximum of the smoothed profile), its density 0.33 ± 0.01 and its temperature 1.15 ± 0.02 of ambient, against 0.54, 0.36 and 1.15 from the continuum model. At $\text{Re} = 32$, where the sonic scale is larger (local Knudsen number at the fitted core width ≤ 0.16), the swirl Mach number of the dense gas (two runs) stays between 0.64 and 0.73 while the target is driven from Mach 0.75 to 3, while the fitted core profile falls from 0.63 to 0.47; the continuum prediction is 0.56–0.60. (A few sparse bins on the axis again read higher, 0.8–1.2, beyond target Mach 3.) At $\text{Re} = 4$ the target reaches only Mach 0.72 and neither description locks. The lock is therefore not an artefact of the closure. What the continuum model does not capture is what happens once the axis has emptied: in every molecular run the core density levels off near a quarter of ambient while the continuum core keeps evacuating, and at $\text{Re} = 16$ beyond target Mach ≈ 1.5 the innermost few σ hold only tens of molecules, with a local mean free path larger than the core. There the smoothed local Mach number reaches 0.84–0.95, while in the surrounding dense gas it overshoots to 0.79 at target Mach 1.5 and then settles at 0.50–0.62. The core thus crosses, within one simulation, from the continuum regime into the free-molecular one — the regime map of §9.4 acting on a single flow. Limitations: a thin slab, with two runs at $\text{Re} = 32$ and three at $\text{Re} = 4$ (target Mach 0.30, 0.50, 0.72: fitted local Mach 0.29 ± 0.02 , 0.40 ± 0.04 , 0.50 ± 0.04 , against 0.27, 0.39, 0.48 predicted beforehand), and a thermostat buffer rather than an open boundary. In a three-dimensional box ($170 \times 170 \times 60 \sigma$, $\text{Re} = 16$) the dense gas stays below local Mach 0.87 while the target passes 3, and the axial density and centroid diagnostics show no fluctuation above those of an undriven vortex in the same box.

The same construction in a Lennard-Jones *liquid* ($\rho = 0.80$, $T = 1$, two runs, $\text{Re} = 4$) tests the cavitation reading of §5. The prediction, again registered before the run, was that the core could not cavitate before its swirl reached the value at which the central pressure deficit $1.70 \rho u_{\text{max}}^2$ equals the ambient pressure ($u_{\text{max}} \approx 1.1$). It cavitated later, at $u_{\text{max}} \approx 1.4$ –1.5, after holding a tension of the order of its ambient pressure; the core density fell from 0.74 to 0.07 of the liquid value within one averaging window and then to vapour density. Once the cavity had opened it grew to a radius of 7.5σ , and the swirl speed at its liquid wall levelled off at 1.77–2.09 (standard error 0.03) while the imposed target rose from 1.6 to 2.8 — at or below the hollow-vortex bound $\sqrt{2p_{\infty}/\rho} = 2.07$ within one standard error in both runs. This is the one genuine *velocity* bound found in this work, and it is supplied by a phase change: a liquid cannot be spun faster than the pressure difference across its vapour core allows. The vapour molecules inside the cavity keep following the force, as the free molecules on the emptied axis of the gas did.

A liquid closer to water ($\rho = 0.79$, $T = 0.75$, ambient pressure 0.27, viscosity 2.95 ± 0.14 ; two slab runs and one three-dimensional box of $110 \times 110 \times 40 \sigma$, all at $\text{Re} = 4$) gave a mixed outcome that we report as it came. The prediction that cavitation would wait for the tensile strength held: the core opened at target swirl 0.95–0.97 in the slab (1.17 in the box), far past the vapour-pressure point at 0.45. The registered prediction of the wall swirl failed: it reached 0.94 in the slab at the half-density radius (1.01–1.02 as the maximum over the adjacent bins) and

1.14 in the box, against 0.82 from $\sqrt{2p_\infty/\rho}$ at the initial pressure (+14% to +24% and +38%). Pressure measured per radial bin (virial tensor plus kinetic part, tested against the global virial) in a further slab run shows why this comparison was unfair: the emptied core pushes liquid outward in the closed periodic box, and the far-field pressure rose from 0.32 to 0.77, so the bound at the concurrent pressure is 1.3–1.4 and the wall swirl stays at 0.45–0.75 of it. This is a post-hoc reading of one run, not a passed test; a closed box cannot test a bound stated for a fixed ambient pressure, and a fixed-pressure boundary is needed. The three-dimensional box shows the same cavitation, and its axial diagnostics are inconclusive for the liquid, since an emptied core has no meaningful centroid and no undriven liquid control was run. An earlier estimate of the wall tension from the cyclostrophic balance started from the initial pressure and is superseded.

9.4 A regime map: which physics a blow-up scenario meets first

Proposition 5.1 is a statement about one route to a singularity. Rereading two others against it — Tao’s blow-up for an averaged Navier–Stokes equation [18] and a recent human-written blow-up for the Euler equations with a force smooth up to the blow-up time [10] — shows that it is one case of something more general. For a flow feature of velocity u and size ℓ , put $\text{Re} = u\ell/\nu$, $\text{Ma} = u/c_s$ and $\text{Kn} = \ell_*/\ell$. Then identically

$$\text{Kn} = \frac{\text{Ma}}{\text{Re}}, \quad \frac{u}{\ell} \leq \frac{c_s^2}{\nu} \iff \text{Ma}^2 \leq \text{Re}, \quad (11)$$

the first being von Kármán’s relation and the second the admissibility bound of Eq. (5) rewritten. Which assumption of the incompressible continuum model fails first is decided by where a scenario travels in the (Re, Ma) plane, and the three constructions travel differently.

The diffusive route ($\text{Re} \approx 1$; the OpenAI Navier–Stokes construction). Here $\text{Kn} = \text{Ma}$: compressibility and rarefaction fail together, at $\ell = \ell_*$. This is Proposition 5.1, and it is the only route on which the kinetic scale is the relevant one.

The inertial route ($\text{Re} \rightarrow \infty$). In Tao’s construction each stage is $(1 + \epsilon_0)$ times smaller and about $(1 + \epsilon_0)^{5/2}$ times faster than the last, and the time between stages is “significantly shorter than the dissipation time at that spatial scale” [18, §1.3]. Velocity therefore grows as $\ell^{-3/2}$ and the stage Reynolds number as $\ell^{-1/2}$, without bound. By Eq. (11) the sonic point is reached at $\ell = \text{Re}\ell_* \gg \ell_*$, where $\text{Kn} = 1/\text{Re} \ll 1$: the first assumption to fail is incompressibility, deep inside the continuum, and the model that takes over is the compressible Navier–Stokes–Fourier system rather than kinetic theory. Any blow-up driven by inertia rather than by a forcing against viscosity — including the boundary scenario of [30, 32] if it were realized in a viscous fluid — is of this kind. Two remarks of Tao’s bear on this paper directly: his mechanism has to be “noise-tolerant” to be built from real fluid, which is the question of Direction 3 (§11.5); and his theorem shows that the energy identity together with harmonic-analysis estimates cannot exclude blow-up, so a regularization that works does so by changing the equation at small scales, not by exploiting structure the equation already has.

The bounded-velocity route (Ma bounded, $\ell \rightarrow 0$). In the forced Euler construction of [10] the circulation and the meridional velocity stay bounded while $\|\nabla\Gamma\|_\infty$ and $\|\omega\|_\infty$ diverge — “the amplitudes themselves can be small while their spatial derivatives increase.” The Mach number never grows, so none of §5 applies. What happens instead is that $\text{Re} = u\ell/\nu$ falls through one at $\ell_\nu = \nu/u = \ell_*/\text{Ma}$, which for a subsonic flow is *larger* than ℓ_* : the neglected viscosity acts before the continuum ends, by the factor $1/\text{Ma}$. This sharpens §7: for Euler constructions of this type the missing physics is viscosity, and the molecular scale is never reached.

So the single scale of Proposition 5.1 is a property of the diffusive route, not of blow-up in general, and the “lock” a real fluid would present depends on the route: kinetic termination on the first, compressibility and heat on the second, plain viscosity on the third. The relations above, the three orderings and the unbounded growth of the stage Reynolds number along

a cascade with Tao’s ratios are formalized in `BlowupRegimeMap.lean` (thirteen declarations, standard axioms).

9.5 A quantum-fluid counterpart: the core that is allowed to be singular

There is one class of fluids in which the microscopic scale is part of the equation of motion, and it supplies an exact counterpart to the diffusive route. Around a singly quantized vortex of a superfluid the velocity is $u = \hbar/(mr)$, so that

$$\frac{ur}{\hbar/m} = 1 \quad \text{at every radius:} \quad (12)$$

with the quantum of circulation $\hbar/m$ in the place of ν , every quantized vortex is a core of Reynolds number one — the condition that defines the construction’s route (§9.4) — by theorem rather than by design. The Gross–Pitaevskii equation is, by the Madelung transform, compressible Euler flow with a barotropic pressure and an additional quantum pressure, and its sonic radius $u = c$ is $\hbar/(mc) = \sqrt{2}\xi$, with ξ the healing length: the counterpart of $\ell_* = \nu/c_s$.

What the quantum fluid does at that scale is instructive. It does not bound the velocity, which diverges on the axis; it removes the mass that would carry it. Solving the stationary vortex, the density vanishes as $\rho \simeq (0.825r)^2$ (units $\hbar = m = c = 1$), reaches half its ambient value at 1.55ξ , and the mass current ρu stays below 0.46. The classical twin — isothermal compressible Euler flow with the same velocity field, whose cyclostrophic density is exactly $\exp(-r_s^2/2r^2)$ — empties its axis too, and the two agree at leading order far from it. A time-dependent simulation of a vortex–antivortex pair makes the same point dynamically: the classical translation speed $\kappa/(2\pi d)$ diverges as the separation $d \rightarrow 0$, whereas in Gross–Pitaevskii every speed we measured is below c — the fastest surviving pair moves at $0.50c$ at a separation of 2.75ξ , closer pairs annihilate into a sound pulse that approaches but does not reach c , and at large separation the speed agrees with the point-vortex law to 0.1% (consistent with Jones and Roberts [13]). Quantum pressure here, heat in the air core of §9.3, and a phase change in water (§5) are three unrelated pieces of microphysics with one outcome: a fluid tolerates $u \propto 1/r$ by evacuating the core. None of them acts as a filter on the velocity, which is a further reason for the withdrawal of §9.6.

Two remarks complete the comparison. First, the superfluid does possess a velocity threshold, and it is a geometric property of the excitation spectrum: Landau’s criterion gives $v_c = \inf_p \varepsilon(p)/p$, the slope of the tangent from the origin to the dispersion curve — exactly c for the Bogoliubov spectrum, zero for a free gas, and about 58 m/s in helium-4, a quarter of the sound speed, because of the roton minimum [14]. A normal fluid has no such threshold, since viscosity dissipates at any speed; what carries over is the method, and the feature of the spectrum that plays the corresponding role in a dilute gas is the termination of the shear mode at $k\lambda = \sqrt{\pi/2}$. That a roton-like collective mode also exists in a normal Fermi liquid [15] indicates that dense liquids keep collective modes down to the interparticle scale, so the gas-kinetic termination should not be carried over to water. Second, combining two published estimates — a minimal kinematic viscosity $\nu_m = \hbar/(4\pi\sqrt{m_e m})$ [16] and a maximal speed of sound $v_u = \alpha c\sqrt{m_e/2m_p}/\sqrt{A}$ [17] — the molecular mass cancels exactly:

$$\ell_* = \frac{\nu}{c_s} \gtrsim \frac{\nu_m}{v_u} = \frac{\sqrt{2}}{4\pi} a_B \approx 6 \text{ pm}, \quad (13)$$

with a_B the Bohr radius, the same for every fluid (water lies a factor 113 above it). The identity is exact; the two inputs are order-of-magnitude bounds, so Eq. (13) is an estimate. The identities of this subsection are formalized in `QuantumVortexLink.lean` (eleven declarations).

9.6 The Leray- α anchor claim, re-examined

Earlier documents of this programme, following advice we recorded verbatim, put the dual-scale intuition in the following form. (i) *Dual scale*: in the Leray- α model the fluid is transported by $\bar{u} = (1 - \alpha^2 \Delta)^{-1} u$, so α is a length above which $\bar{u} \approx u$ and below which the filter engages. (ii) *Lock*: the three-dimensional Leray- α equations have global smooth solutions, so “the singularity is mathematically impossible” and α “acts as an absolute lock on gradient growth.” (iii) *Physical anchor*: setting $\alpha = \ell_*$ makes Leray- α “a mathematically sound representation of the fluid hitting its continuum limit.” We can now say precisely what survives.

Claim (i) is a correct description of the model. Claim (ii) is a correct theorem [35, 40] about a different equation: it says nothing about Navier–Stokes, and in the light of [18] it could not, since it is obtained by weakening the nonlinearity by two derivatives at small scales — by leaving the supercritical class, not by using finer structure. It is also not a lock on the constructions at issue: on a $\text{Re} \approx 1$ forced collapse the filter changes the core size by 0.2–0.4% (§9.2), and on the pure swirl column its nonlinearity is a gradient and does nothing at all.

Claim (iii) is false, for four reasons of increasing generality.

1. *The filter width does not appear in the linear dynamics.* Leray- α and LANS- α alter only the quadratic term, and the derivative at rest of $u \mapsto B(F_\alpha u, u)$ vanishes for every bounded bilinear B and every bounded linear F_α (`LerayAlphaLinearization.lean`). Small disturbances therefore relax in Leray- α exactly as in Navier–Stokes, at the rate νk^2 , for every α . In the gas they relax at $\nu k^2[1 - (k\lambda)^2 + \dots]$, never faster than $1/\tau$, and not at all beyond $k\lambda = \sqrt{\pi/2}$ (Eqs. 9–10); νk^2 exceeds any such cap. No value of α reconciles the two, because the part of the dynamics the continuum limit changes first does not contain α .
2. *It regularizes in the wrong direction.* The filter removes small-scale transport and leaves small-scale damping intact; kinetic theory leaves transport and *reduces* small-scale damping (the Burnett coefficient is +1).
3. *What the continuum limit brings is not a closure of the same equations.* At ℓ_* the Mach, Knudsen and Eckert numbers are all of order one (Proposition 5.1). A model that remains incompressible, isothermal and Newtonian at that scale represents none of the three effects, whatever it does to the transport velocity; §9.3 measures two of them.
4. *ℓ_* is the relevant scale on one route only* (§9.4). On the inertial route the model fails at $\text{Re } \ell_*$ by compressibility, and on the bounded-velocity route at ℓ_*/Ma by viscosity; a filter at ℓ_* is placed below both.

None of this detracts from Leray- α as a turbulence closure at scales far above ℓ_* , which is what it was designed for and where its global regularity is an asset. What is withdrawn is the identification of its filter width with the scale at which the continuum ends. The dual-scale intuition itself survives in a different form, which is the content of Eq. (11): there are two scales in every blow-up scenario, the scale of the flow and the scale ℓ_* fixed by the molecules, and their ratio is Ma/Re . The only unconditional bound in the hierarchy of models is at its bottom: N particles of mass m with kinetic energy at most E have speeds at most $\sqrt{2E/m}$ (formalized in the same file), and no continuum or kinetic equation inherits it.

10 Discussion

The OpenAI system’s achievement is a genuine milestone in automated reasoning. It exploited the analytical gap between C^∞ and real-analytic functions, using Gevrey-class cutoffs to construct a mathematically flawless singularity that the Lean 4 kernel rigorously certifies. It did not cheat; it solved the mathematical problem as posed.

Five observations, taken together, show that the constructed singularity exists within the abstract mathematical continuum $\mathbb{R}^3$ without describing anything a real, physical fluid does: (1) intensive local energy density and enstrophy diverge ($\sim \tau^{-1.010}$, $\sim \tau^{-0.515}$) while the global energy vanishes; (2) compressibility, rarefaction and viscous heating all become order-one at a single scale $\ell_* = \nu/c_s$ — a few picoseconds before blow-up in water, nanoseconds in air, essentially independently of the initial vortex size, a coincidence that belongs to the diffusive route this construction takes (§9.4) — and in a liquid cavitation intervenes three decades earlier still; (3) the moment-matching system’s apparent ill-conditioning is a non-dimensionalization artifact, not physical fragility; (4) the force is backward-specified and has no independent physical origin; and (5) the Euler datum needs coherent structure below the molecular length, where the missing physics is viscosity.

This is not, we want to stress, a novel or contested position among fluid dynamicists: it is close to the median reaction from the community discussion summarized in §11, and it echoes a long-standing point about continuum models generally — Ohm’s law, ideal-gas equations of state, and other workhorse relations all have known breakdown regimes without this being considered a mark against their mathematical content. Three implications follow:

1. **For mathematics:** The proof is definitive. Alternatives C and D are resolved as statements about the continuum incompressible model.
2. **For physics:** This particular singularity requires the fluid to remain a continuum, incompressible, isothermal medium down to ℓ_* — a few molecular diameters in water — and, in a liquid, to sustain tensions of order 10^2 MPa for picoseconds. It is not an observable. What *is* observable is the question of how a real fluid leaves the model: cavitation, compressibility, rarefaction and heating, in an order that Proposition 5.1 makes explicit. None of the mechanisms we could test — kinetic relaxation, compressibility, viscous heating — stops a driven $\text{Re} \approx 1$ core before ℓ_* (§9); what a real fluid presents instead depends on the route to the singularity.
3. **For AI-assisted mathematics:** A kernel-checked proof of a physically motivated equation is not automatically a physical result, and the gap between the two is worth making explicit — and, where possible, checkable — as part of the proof pipeline (§11.3).

A fourth implication concerns this paper’s own method, and §9 is the occasion for it. Having spent the preceding sections establishing where a construction stops describing a fluid, we proposed a remedy — a cutoff regularization — and then found, on testing it, that the remedy could not be evaluated at all with generic data, because it presumes the very core structure that generic data does not produce. The asymmetry is worth naming: it is easier to identify where a model fails than to establish that a proposed repair does anything, and the standard of evidence we applied to OpenAI’s construction is the one we then had to apply to our own suggestion. That the suggestion survived only as a conditional statement, with its premise promoted from an implicit assumption to the object actually under test — and that its most ambitious form, a Leray- α filter of width ℓ_* as a representation of the fluid at its continuum limit, did not survive at all (§9.6) — seems to us the useful outcome rather than a disappointing one.

11 Outlook: Repositioning, Community Directions, and Physically-Grounded AI

11.1 Mathematics at Its Limit, Not a Claim About Physics

Section 1 already previewed the position we think the evidence supports, and which was arrived at independently by much of the fluid-dynamics community when an early version of this audit was first posted for feedback on `r/FluidMechanics` [6] (participants acknowledged individually

in Acknowledgments): the Millennium Prize problems, Navier–Stokes existence and smoothness included, were set as precise tests of mathematical technique against a specific, well-defined continuum model — not as referenda on whether that model perfectly describes real fluids. The consensus of that discussion, repeated in different words by several commenters, was that Navier–Stokes “has always been a useful approximation and not fully physical,” comparable to other continuum relations (Ohm’s law was cited) with known breakdown regimes; that a blow-up result mainly tells us where the model runs out of validity rather than anything about real fluids; and, on the specific question of whether the incompressibility breakdown identified in §5 invalidates the OpenAI proof, that it does not — in the words of one exchange, the incompressible approximation fails to capture what a real fluid does, but *within* the (non-physical) incompressible world the OpenAI proof remains valid. The full quotations, with usernames, are preserved in Appendix B for the historical record.

We agree with that statement without qualification. Everything in §2–§7 should be read accordingly: not as claims that the Lean proof is wrong, but as a map of exactly where and how far this specific constructed solution departs from the regime the incompressible model is normally trusted to describe. That map is, we think, the useful residue of this kind of audit, and several commenters independently proposed uses for it that go well beyond simply flagging unphysicality, summarized next.

11.2 Research Directions from the Community Response

Four concrete directions recurred in that discussion and align with work streams already underway in this project’s companion directory, `05_Community_Research_Directions/`, which credits the same discussion:

1. **An admissibility, or entropy, condition for 3D Navier–Stokes.** Part of the discussion proposed that 3D Navier–Stokes may need an augmented admissibility condition, analogous to the Lax entropy condition used to select the physically correct weak solution of the inviscid Burgers equation, to filter out unphysical weak solutions such as this one — the blow-up result tells us that the equations, unaugmented, allow something a physical theory should not; a related suggestion was that the Prize problem itself might need restatement so that it pertains only to physical fluids. This direction is developed in `WorkStream1_EntropyCondition`.
2. **The pre-singularity trajectory as a turbulence-modeling resource.** Other comments noted that even though the endpoint is unphysical, the approach to it — before the Mach or Knudsen thresholds in Table 6 are crossed — may still carry information relevant to subgrid-scale closure modeling (LES/RANS). This direction is developed in `WorkStream3_Turbulence`.
3. **Deriving, rather than assuming, the thermal response.** A further comment correctly pointed out that incompressible Navier–Stokes has no energy equation, so any thermal-shock argument built on top of it (including ours in §4) is an external estimate, not a derived consequence. A genuine energy/temperature-coupled extension, or a cavitation/boiling model, is needed before the thermal argument can be sharpened past an order-of-magnitude check. This direction is developed in `WorkStream2_ThermalEquation` and `WorkStream4_MechanicalCavitation`; a first quantitative step, a forced core in a compressible fluid with an energy equation, is reported in §9.3.
4. **Teaching proof-search AI systems basic thermodynamics.** The most actionable suggestion distilled from the discussion was, in essence, to teach AI proof assistants basic thermodynamics so they catch these edge cases. That suggestion directly motivates §11.3.

Full attribution for each of these points is given in Appendix B.

None of these four directions is resolved by this paper (the third has a first step, the fourth a partial implementation, below); we record them as open invitations, consistent with the standing call for contributions in `05_Community_Research_Directions/README.md`.

11.3 Toward Physically-Grounded AI Theorem Proving

Motivated directly by the fourth point above, we propose a three-pillar architecture for future AI-assisted formalization of physically motivated equations, restating an earlier internal draft manifesto (`TAO_NEUROSymbolic_Scientific_AI_Manifesto.md`) in a form scoped to what is realistically buildable rather than what is aspirational:

1. **Neural proposal.** An LLM-based multi-agent system, as in OpenAI’s own architecture, proposes candidate constructions, tactics, and similarity profiles.
2. **Symbolic verification.** A Lean 4 (or equivalent) kernel checks each candidate for syntactic and logical correctness — exactly as today, with zero `sorry` and zero unauthorized axioms. This pillar alone is what makes a result a *theorem*; nothing below changes that.
3. **Physical admissibility gate.** A third, independent checker evaluates the *same* construction against an explicit, falsifiable list of physical admissibility predicates before it is reported as *physically* relevant (not before it is accepted as a mathematical theorem). For incompressible Navier–Stokes the list is short and dimensionally derived (§5.7): the local vorticity bound $|\omega| \lesssim c_s^2/\nu$ of Eq. (5), which encodes $\text{Ma} \lesssim 1$ and $\text{Kn} \lesssim 1$ at once (equivalently $\text{Ma}^2 \leq \text{Re}$, §9.4); for liquids, a core pressure deficit ρu^2 below the tensile strength; and, where an energy equation is present, an Eckert number $u^2/(c_p \Delta T)$ that stays small. Each predicate is a scaling estimate with $O(1)$ prefactors, and should be reported as such.

A rational-arithmetic reflection bridge (`openai_poc_pi_verifier.py`) already implements a narrow version of pillar 3’s Mach/Knudsen checks in Lean, using exact rational arithmetic evaluated via `#eval`; it is a reasonable starting point for the broader gate above, though it does not reach into the OpenAI proof objects themselves — the Lean statement of Direction 2 does.

The same gate is also useful outside theorem proving, and we have implemented it as a runtime monitor in the solver of §9: each step, the flow is checked against Eq. (5) and the Mach and Knudsen thresholds, in the fluid’s own physical units, and the first crossing of each is reported. The point is a distinction that numerical practice tends to blur: a simulation can be perfectly stable and still describe a state no real fluid can occupy. Running the same decaying flow under two physical interpretations makes this concrete — read as 1 m/s across 1 cm of water it stays nine orders of magnitude inside the bound, and read as 300 m/s across 10 nm it reaches $\text{Ma} = 0.20$, $\text{Kn} = 0.05$ and $|\omega|/(c_s^2/\nu) = 0.05$ — with byte-identical numerics in both cases. Model validity is not a property of the discretization, and a solver that reports its own departure from the modelled regime seems to us more useful than one that regularizes the departure away.

We want to be explicit about the current state of this project’s own attempt at pillar 3. Ten Lean files in this repository compile with no `sorry` and no axioms beyond the standard three, together 85 declarations checked with `#print axioms`. One states results about OpenAI’s *actual* objects: it imports their `ProblemStatement` and periodic-integration library unchanged, is described under Direction 2 of §11.5, and assumes nothing beyond their definitions. Four support the physical reading: the scaling chain of Proposition 5.1 as an equality chain from the diffusive core scalings (core Reynolds number one, the Mach and Knudsen times, Eckert number one, and their coincidence at ℓ_*); the regime map of blow-up scenarios of §9.4; and two statements about the kinetic model — the H-theorem, conservation and positivity of the discrete-velocity BGK step used in §9.2, and the exact cap $-1/\tau \leq \text{Re } \mu \leq 0$ on the eigenvalues of a skew streaming

operator plus projected relaxation. Four concern the regularizations: the bounds on the Leray- α filter symbol; an H-theorem for a simpler discrete BGK collision step; the finite-dimensional identity behind the gate–drain distinction of §9.2, that a projected skew-adjoint term changes the energy only at second order in the step while a dissipative one removes it at first order; and the absence of the filter width from the linearization of every α -model, which is the formal reason given in §9.6. The tenth collects the exact identities of the quantum-fluid comparison of §9.5. Each file’s header states what it does not prove.

11.4 The Dual-Scale Cutoff Hypothesis: History and Outcome

Earlier versions of this paper stated one further-out idea, separately from the buildable architecture above, as a hypothesis for future investigation. We record its history and outcome here rather than as a live proposal. Earlier drafts of this project proposed a “dual-scale” effective wavenumber saturating at large $|k|$,

$$k_{\text{eff}}(k) = \min \left(|k|, \frac{1}{\alpha' |k|} \right) \quad (k \neq 0), \quad k_{\text{eff}} \leq 1/\sqrt{\alpha'}, \quad (14)$$

with $\alpha' = \ell_{\text{min}}^2$, and motivated it by string-theoretic T-duality and Trans-Planckian censorship [46]. The inequality is true and Lean-checkable (`dual_scale_wavenumber_bounded` in this repository), but it is an algebraic fact about a formula, and the string-theoretic motivation adds nothing: stripped of that language, a regularization of this kind is a member of the Leray- α / Bessel-filter family $\bar{u} = (1 - \alpha' \Delta)^{-2} u$, whose global regularity is known [35, 40]. The draft that asserted a substantive physical link, via a $K3 \times T^2$ compactification, was reviewed and its physical claim withdrawn (its central Lean theorem ended in a bare `True`, and its vorticity equation lacked a term); it now carries a withdrawal notice in this repository, and the companion note [5] records the corrections.

What survived that review was a hypothesis with a definite test — the cutoff law $u_{\text{max}} \sim \nu/\sqrt{\alpha'}$ — which earlier versions recorded as not yet performed. It has since been performed (§9). The law is exact given a $\text{Re} \approx 1$ core and untestable without one; on a manufactured core a dissipative barrier stalls the collapse near $\sqrt{\alpha'}$ without a converged arrest; and the reading of the cutoff length as the scale at which a real fluid leaves the continuum is withdrawn (§9.6). What remains of the dual-scale idea is the relation $\text{Kn} = \text{Ma}/\text{Re}$ between the flow’s scale and the molecular one (§9.4). The research directions that follow are set out in §11.5.

11.5 Three directions, in detail

The community suggestions of §11 are broad. The three directions below are the ones we think are both consequential and startable now, and we give each in enough detail to be acted on — including the failure modes, since a direction stated without its failure modes is an advertisement rather than a plan. Four shorter items follow.

Direction 1. A forced-core test bed for cutoff regularizations. The first two stages of this direction are now built and reported in §9.1 and §9.2: a manufactured $\text{Re} \approx 1$ collapse tracked to 10^{-7} , the single-variable collapse of the barrier’s engagement, and the gate–drain comparison. Four things remain, in order of value. First, a converged arrest: at 96^3 the core stalls near $(1.2\text{--}1.5)\sqrt{\alpha'}$ with exponents approaching the law but not yet settled, and $B/F = 1$ proved the wrong detector (§9.1); a longer resolved window is needed. Second, a collapsing *axial* scale: the meridional cell used here has ℓ_z fixed by the box, whereas the construction has $\ell_z \asymp \tau^{1/2-h}$, which needs a localized axial envelope. Third, the azimuthal stability of the forced column, which the axisymmetric targets used so far cannot probe and which could destroy the core for reasons unrelated to any regularization. Fourth, the construction’s own profile with $h > 0$ in place of the Gaussian surrogate. One of the failure modes this direction was designed

to catch — the forcing doing the work the dynamics should do — is now measured rather than merely anticipated: the B/F diagnostic is exactly that ratio. The same target has since been used to drive a kinetic gas and a compressible, heat-conducting one (§§9.2–9.3).

Direction 2. Sharpening the admissibility condition. The bound $|\omega| \lesssim c_s^2/\nu$ of §5.7 is dimensionally forced but blunt. Its candidates can be ordered by how finely they discriminate: a bound on vorticity *magnitude* (the present one); a bound on *transfer* across a scale, which is what Leray- α enforces and which §9 measured to be energy-conserving rather than dissipative; and a bound on vorticity *direction*. The middle term is where the dual-scale intuition behind earlier drafts of this programme was placed, and §9.2 now limits its reach: a transfer bound is the effective constraint only at $\text{Re} \gtrsim 10\text{--}100$, and at ℓ_* , where $\text{Re}_r = O(1)$, it is the weakest of the three; §9.6 withdraws its identification with the continuum limit. Three concrete questions follow.

Wrong field? Viscous dissipation density is $2\nu|S|^2$ with S the rate-of-strain tensor, not $\nu|\omega|^2$. The two agree in integral ($\int 2|S|^2 = \int |\omega|^2$ on a domain without boundary) and in order of magnitude, but not pointwise — and every physical mechanism in §5, heating especially, is driven by the local dissipation. A criterion stated on $|S|$ would be the physically correct object; it is stated on $|\omega|$ here because Beale–Kato–Majda is.

Wrong functional? Magnitude may be the wrong thing to bound at all. Constantin and Fefferman [42] showed that regularity follows from sufficient *coherence of the vorticity direction* in regions of large vorticity, rather than from any bound on its magnitude. That is strikingly close to the property this paper has repeatedly identified as the construction’s essential and physically fragile ingredient — coherent phase alignment maintained down to arbitrarily fine scales (§7). A direction-based admissibility criterion would therefore discriminate exactly where a magnitude-based one does not, and would connect the geometric fragility to a regularity theorem rather than to an estimate. We regard this as the most promising single item on this list.

Formalized against the construction — and what that turned out to require. The narrow step proposed in earlier versions is now done. Admissibility is stated on OpenAI’s own `VelocityField`, as a bound on the operator norm of `spatialDerivative`; that norm dominates both strain and vorticity, which settles the “wrong field?” question for the purposes of a formal statement. Unconditionally, any object satisfying OpenAI’s `CandidateProperties` has a uniformly bounded gradient on every $[0, 1 - \delta]$, and — since a uniformly bounded gradient on $[0, 1)$ excludes `SpeedUnboundedAtOne`, proved below — the same object exceeds every gradient bound in every window before $t = 1$, and therefore leaves the admissible set for every fluid and every choice of units. All results compile inside OpenAI’s own project with only the standard axioms and no remaining hypothesis.

Earlier versions said this hypothesis would be the Beale–Kato–Majda theorem. It is much weaker, and saying so matters. OpenAI’s candidate properties include spatial periodicity of velocity, pressure and force, and zero initial velocity. On the periodic cell the pressure gradient, the viscous term and the divergence-free advection all average to zero, so the cell mean obeys $d\langle u \rangle/dt = \langle f \rangle$ and stays bounded; a gradient bound C then confines u to within $\sqrt{3}C$ of that mean. The hypothesis is an elementary lemma. Earlier versions left it unformalized for want of integration by parts over the periodic cell; OpenAI’s own project supplies exactly that (`NavierStokes.PeriodicIntegration`), and with it the lemma is now proved, so the formal result is unconditional. The honest reading follows: the formal result is close to “velocity blow-up forces gradient blow-up,” and its value is that it is the first statement in this project about the proof objects themselves, not that it adds physics.

Finally, a limit worth recording: an admissibility condition that excludes blow-up is weaker than one that selects a unique physical solution. Non-uniqueness of Leray solutions of the forced equations [34] is not addressed by any bound of this kind, and we do not claim it is.

Direction 3. Coherent pulses under thermal noise. The solver of §9 includes a Landau–Lifshitz noise term calibrated by the fluctuation–dissipation theorem, and one data point has been taken with it (reported with the experiment files, not in this paper). The data point is not yet the question, because the quantity that matters is coherence rather than amplitude: the construction’s pulses are phase-locked and shear-amplified in a specific arrangement, and a random perturbation of equal energy is a different object. The sharp question is a competition between two rates — the shear amplification rate σ supplied by the background, and the rate at which noise destroys the phase relation the amplification depends on. Amplification survives when $\sigma \tau_{\text{decoh}} \gg 1$, and τ_{decoh} is computable from the noise variance at the packet’s wavenumber, so the criterion is a number, not a judgement. The natural amplitude scale to compare against is the thermal length $\ell_{\text{th}} = k_B T / (\rho \nu^2)$, which is picometres in water, well below the molecular spacing; the comparison that matters is therefore made at the scale where amplification begins, not at the scale where the construction ends.

A first attempt clarifies what the measurement requires. A phase-definite single-mode packet seeded on a Taylor–Green background is not amplified in any orientation we tried: its amplitude decays several times faster than viscosity alone, by scattering to neighbouring wavenumbers. With $\sigma < 0$ the competition the criterion describes cannot arise, so the result is uninformative rather than negative. It needs a *localized* packet in a region of sustained strain, which is what the forced core of Direction 1 provides, and a packet amplitude above the thermal floor at its wavenumber from the start — the attempt began below it.

What makes this worth doing is the strength of the conclusion available. Every argument in §5 concerns the approach to the singularity: the model fails at ℓ_* , some picoseconds before $t = 1$. A decoherence result would instead concern $t = 0$ — it would say the required datum cannot be prepared in a fluid at finite temperature, which is a statement about realizability rather than about a limit reached en route. The caveat is symmetric and must be kept in view: fluctuating hydrodynamics is itself a coarse-grained model with a validity range, and pushing it to nanometre scales repeats, in a new place, exactly the error this audit was written to correct.

Four shorter items. *The molecular test, done in a slab.* The closure-free test of the Mach lock proposed in earlier versions has been carried out by molecular dynamics (§9.3): it confirms the lock where the gas is a continuum, and in a liquid it finds cavitation followed by a capped wall speed. Three-dimensional boxes, further runs at $\text{Re} = 4$ and 32 and a water-like liquid have now been run (§9.3); what remains is a fixed-pressure far field for the liquid bound, an undriven liquid control for the axial diagnostic, and an ensemble larger than one to three runs per case. *Codimension.* The number of unstable modes of the linearized self-similar profile beyond those generated by symmetries is the rigorous form of “non-generic”; it has been computed for related Euler and Boussinesq profiles by computer-assisted and neural-network methods [32, 33], and needs eigenvalue analysis in similarity variables rather than the forward integration used here. A finite count would not make the blow-up physically irrelevant, since unstable self-similar solutions still organize transient dynamics. *Dyadic criticality.* In checking whether our exponents meet a formalized shell-model threshold, we found that the construction’s enstrophy exponent at $h \rightarrow 0$ coincides with the rate saturating the dyadic estimate that carries the same scaling as three-dimensional Navier–Stokes. We record it as a numerical coincidence, not a theorem: the shell model’s single length scale is an obstruction to deriving it, given $\ell_r \neq \ell_z$ here. It is also independent of, and pulls opposite to, the Serrin-threshold marginality of §4. *Validity monitoring in practice.* The monitor of §11.3 is a small amount of code and would cost little in a production solver; whether reporting departure from the modelled regime changes anything about how such solvers are used is an empirical question about practice, not about fluids.

12 Conclusion

OpenAI’s formalization is a genuine, kernel-verified mathematical achievement, and Alternatives C and D are, as mathematical statements about the continuum incompressible model, resolved by it. Nothing in this paper disputes that, and nothing in it could: the Lean kernel already settled the question of syntactic correctness before we started.

What this paper adds is a physical reading: a scaling table for locally divergent quantities that respects the Prize’s global energy bound; a unit-free validity scale $\ell_* = \nu/c_s$ at which compressibility, rarefaction and heating enter together; the observation that in liquids cavitation comes first; a single local admissibility condition $|\omega| \lesssim c_s^2/\nu$ that Beale–Kato–Majda turns into a theorem; a demonstration that the moment-matching system’s apparent ill-conditioning is a non-dimensionalization artifact, and that the radiated-acoustic-energy check is the Mach criterion in disguise; an honest accounting of the backward-specified force; a description of the Euler datum’s dependence on coherent structure below the molecular length; and a Lean 4 proof, on OpenAI’s own definitions and with no additional hypothesis, that any object with their candidate properties leaves every such admissible set near the singular time. None of these observations makes the mathematics wrong, and none of them, on their own, proves that real turbulent fluids cannot develop extreme, short-lived, localized events resembling a scaled-down piece of this construction — that remains a genuinely open empirical and theoretical question, not one this audit resolves. (Readers of earlier public drafts of this audit will find a full account of what changed, and why, in Appendix A.) What they do establish is that *this particular* construction is not a description of anything a real fluid does; it is a demonstration of what the continuum model, pushed to the exact edge of its own internal consistency, is mathematically capable of.

We also tested, rather than merely proposed, the one repair this paper had to offer. The cutoff regularization of §11 turns out to be exact given its premise and untestable without it: a $\text{Re} \approx 1$ collapsing core is a property of the construction, not of flows in general, so no amount of resolution applied to generic data can decide the question (§9). We then built the premise and measured on it (§9.1, §9.2). The barrier’s engagement reduces to a function of one variable, and transport-filtering regularizations — the family in which the intuition behind earlier drafts was placed — turn out to be one to two orders of magnitude weaker than dissipation on exactly the kind of collapse that reaches ℓ_* , because there they can act only through a nonlinearity that is a few percent of the dynamics. Nor does the physics beneath the continuum behave like the stronger of the two: the validity scale is the mean free path, with a constant derived from kinetic theory, and the exact kinetic shear spectrum damps small scales less than viscosity, caps the rate at the collision frequency, and ends the hydrodynamic mode at $k\lambda = \sqrt{\pi/2}$ — and a nonlinear kinetic simulation of the forced collapse finds no arrest there; nor, on this route, do compressibility and heat stop the core before ℓ_* . What acts where the continuum ends is the end of the hydrodynamic description, which is where this paper began. That conclusion belongs to the diffusive route the construction takes. On an inertial route the first physics to arrive is compressibility and heat, and there a real gas does lock — in Mach number, at about 0.7, not in velocity or size (§9.3, §9.4); on a bounded-velocity route it is plain viscosity. The identification of a Leray- α filter width with ℓ_* is withdrawn (§9.6). The directions of §11.5 follow, and the second of them — an admissibility criterion stated on the coherence of vorticity direction rather than on its magnitude, after Constantin and Fefferman [42] — still strikes us as the most likely to say something that is both true and new, because it bears directly on the coherence the construction requires and the thermal environment denies.

The more useful outcome of this exercise, in our view, is not the verdict but the map: a concrete list of where the physical assumptions break down, a set of research directions raised independently by the community that received this audit (§11), a proposed — and only partially built — architecture for keeping future AI-generated proofs of physical PDEs honestly labeled

as mathematical or physical results, and the part of the dual-scale idea that survived testing — the ratio Ma/Re between the flow’s scale and the molecular one — for anyone who wants to push it further than we have. The AI solved the mathematician’s problem. Whether, and how, that result connects to the physicist’s problem is still an open question, and, we think, a more interesting one.

Acknowledgments

This paper was substantially improved by an open peer review received 15 September 2026, which recommended, among other changes, that specific Reddit pseudonyms be moved out of the main text and into this section, and that differences from earlier public drafts be consolidated rather than scattered through the text; both recommendations are implemented here and in Appendix A. We also thank the participants of the `r/FluidMechanics` discussion of an early version of this audit [6], whose contributions are quoted in full in Appendix B: `Leodip`, `mrbiguri`, `Dihedralman`, `brokebeany`, `Fluid.Substance5132`, `cowgod42`, `tlmbot`, `babainottawa`, `Cute.Presence.2861`, and `vibe0009`. Their objections and suggestions are the direct origin of §11 and of the `05.Community.Research.Directions` work streams.

A Note on Previous Versions

This paper has circulated in several public drafts since its first posting. In the interest of a main text that stands on its own, differences from those drafts are consolidated here rather than noted inline.

Claims withdrawn. Three prior claims do not appear in this version and should be treated as withdrawn, not merely superseded: (i) that the constructed singularity reaches “plasma temperatures” (the correct estimate, $\Delta T \simeq u^2/c_p$, gives hundreds of kelvin at the Mach limit in water, not 10^4 K; §5); (ii) that a Lighthill acoustic radiative-energy check independently corroborates the Mach breakdown time (it is dimensionally Ma^5 , i.e. the Mach criterion restated, not an independent check; §5); (iii) that the moment-matching Jacobian’s raw condition number $\kappa \sim 10^{28}$ measures physical fine-tuning or fragility of the singularity (it is an artifact of unscaled physical dimensions in the matrix entries, $\kappa(B) \approx 4.1 \times 10^5$ once non-dimensionalized; §3). A fourth item, a “bounded-ensrophy” admissibility axiom stated as a global integral bound with a numerical constant $\Omega_{\max} \approx 1.13 \times 10^{13} \text{ s}^{-2}$, has been replaced outright by the local, dimensionally derived bound of Eq. (5); the global form is not used in this paper.

Material removed. Two figures — a Taylor–Green-vortex comparison whose resolution and normalization were not documented to a reproducible standard, and a spectrum plot in which the “OpenAI” curve was a hand-drawn schematic rather than a computed result — are not reproduced here; see §8.

Citations corrected. The moment-matching system referenced in §3 is Proposition B.8 and the moment conditions (5.10)–(5.11) of [2], not “Lemma 8.7,” which is a distinct lemma addressing a different step of the construction. The figure of “10,000 agents” collaborating for “88 hours,” which circulated in secondary press coverage, is not stated in either OpenAI paper and is not repeated here as fact (§1).

Reframed, not retracted. The “dual-scale” effective-wavenumber cutoff of §11 was previously motivated by an asserted physical connection to string-theoretic T-duality; that physical claim was independently reviewed and withdrawn (the relevant Lean theorem’s load-bearing conjunct

was a bare `True` placeholder, and its supporting vorticity-transport equation was missing a term). The mathematical content — a saturating regularization in the Leray- α family with a testable cutoff law — survives with the physical motivation removed; see [5] for the full account of what was found and corrected.

Withdrawn in version 5.5.0. Documents of this programme, and the framing of §9 in earlier versions, entertained the identification of a Leray- α filter width with the validity scale ℓ_* . §9.6 withdraws it and gives the reasons. A quasi-steady prediction made during this version’s work, that the density hole would arrest the forced core at $\text{Ma} = 0.36$, also failed and is reported as such in §9.3.

Forecasts not borne out. Version 5.3.0 stated that a 96^3 forced-core sweep would reach the $B/F = 1$ crossing for the upper half of its band. It reached it for none; the result and the reason are reported in §9.1. The same version described the admissibility result of Direction 2 as conditional on an unformalized lemma; that lemma is now proved. Version 5.4.0 quoted one grid-refinement endpoint of the kinetic collapse (§9.2) as 0.86λ , a value of a different metric; with the metric used for the other endpoints it is 0.96λ , corrected here, with the conclusion unchanged. Earlier versions also quoted Lean counts (45 theorems) under a different counting convention; this version counts declarations checked with `#print axioms`.

B The `r/FluidMechanics` Discussion, in Full

An early version of this audit was posted for open discussion on `r/FluidMechanics` by one of the present authors [6]. The exchange materially shaped this paper (§11), and the full quotations, with the usernames under which they were publicly posted, are preserved here.

On whether the Millennium Prize problems were ever claims about physical fluids, `Leodip` wrote: “the point of the millennium prize problem was never to figure out anything with regards to real world fluids... finite-time blowup is totally unphysical either way and just a result of the N-S model.” `mrbiguri` added that a blow-up result “suggests that it’s not fully descriptive of real world physics,” and `Dihedralman` placed this alongside other familiar breakdowns of continuum models: “Ohm’s law breaks down... Navier Stokes has always been a useful approximation and not fully physical.”

Asked by `brokebeany` whether the incompressibility breakdown identified in §5 invalidates the OpenAI proof, `Fluid_Substance5132` gave what we think is the cleanest statement of the right frame: “In the real world[,] the incompressible approximation fails to capture what actually happens. Within the (non physical) incompressible world[,] the o[pen]ai proof remains valid.” We agree with that statement without qualification.

On what should replace a bare unphysicality verdict, `cowgod42` proposed that 3D Navier–Stokes may need an augmented admissibility condition, analogous to the Lax entropy condition used to select the physically correct weak solution of the inviscid Burgers equation: “the ‘correct’ equations of fluids should not allow this to happen... we need to change the equations, or augment them in some way.” `tlmbot` independently suggested that the Prize problem itself might need restatement so that it pertains only to physical fluids. `babainottawa` and `Cute_Presence_2861` both noted that even though the endpoint is unphysical, the pre-singularity trajectory — before the Mach or Knudsen thresholds of Table 6 are crossed — may still carry information relevant to subgrid-scale closure modeling. `vibe0009` correctly pointed out that incompressible Navier–Stokes has no energy equation, so any thermal-shock argument built on top of it, including an earlier version of ours, is an external estimate rather than a derived consequence — the point that motivated the sharper treatment of §5 in this version.

In a follow-up post summarizing the initial round of feedback, the audit’s authors distilled the most actionable suggestion as teaching AI proof assistants basic thermodynamics so that

they catch edge cases like this one; that suggestion is the direct motivation for §11.3.

References

- [1] C. L. Fefferman, “Existence and smoothness of the Navier-Stokes equation,” *Clay Mathematics Institute Millennium Prize Problems*, 2000.
- [2] OpenAI, “Finite time blowup for Navier-Stokes and Euler equations,” Lean 4 Formalization, github.com/openai/NavierStokesAndEuler, 2026.
- [3] OpenAI, “On the Navier–Stokes Millennium Prize Problem,” September 8, 2026, <https://openai.com/index/navier-stokes-solution/>.
- [4] OpenAI, “Finite time blowup for the Euler equation,” preprint and Lean 4 formalization, github.com/openai/NavierStokesAndEuler, 2026.
- [5] X. Callens, “Where the continuum ends: the OpenAI Navier–Stokes blow-up as a probe of model validity, two-scale closure, and formalized physics,” version 2, this repository (paper/where_the_continuum_ends.tex), September 2026.
- [6] r/FluidMechanics, “The fluid dynamics behind OpenAI’s Navier-Stokes proof: Why the fluid would vaporize before the singularity” and follow-up discussion, thread by u/OrganizationTop9026, September 2026, <https://www.reddit.com/r/FluidMechanics/comments/1wekxy3/>.
- [7] L. Escauriaza, G. Seregin, V. Šverák, “ $L_{3,\infty}$ -solutions of Navier-Stokes equations and backward uniqueness,” *Russian Math. Surveys* **58**(2), 2003.
- [8] J. T. Beale, T. Kato, A. Majda, “Remarks on the breakdown of smooth solutions for the 3-D Euler equations,” *Comm. Math. Phys.* **94**, 61–66, 1984.
- [9] R. Penrose, “Gravitational collapse: The role of general relativity,” *Riv. Nuovo Cimento* **1**, 252–276, 1969.
- [10] “Blowup for the Euler equations with smooth forcing,” manuscript (2026), companion to L. Alpöge and T. Buckmaster, “Blowup for the Boussinesq equations with smooth forcing”; <https://cims.nyu.edu/~tristanb/euler.pdf> (consulted 17 September 2026; the copy consulted carries no author line).
- [11] F. Merle, P. Raphaël, I. Rodnianski, and J. Szeftel, “On the implosion of a compressible fluid II: Singularity formation,” *Ann. of Math.* **196** (2022), 779–889.
- [12] T. Buckmaster, G. Cao-Labora, and J. Gómez-Serrano, “Smooth imploding solutions for 3D compressible fluids,” *Forum Math. Pi* **13** (2025), e6.
- [13] C. A. Jones and P. H. Roberts, “Motions in a Bose condensate: IV. Axisymmetric solitary waves,” *J. Phys. A* **15** (1982), 2599–2619.
- [14] H. Godfrin and E. Krotscheck, “The dynamics of quantum fluids,” arXiv:2206.06039 (2022).
- [15] H. Godfrin, M. Meschke, H.-J. Lauter, A. Sultan, H. M. Böhm, E. Krotscheck, and M. Panholzer, “Observation of a roton collective mode in a two-dimensional Fermi liquid,” *Nature* **483** (2012), 576–579.
- [16] K. Trachenko and V. V. Brazhkin, “Minimal quantum viscosity from fundamental physical constants,” *Sci. Adv.* **6** (2020), eaba3747.

- [17] K. Trachenko, B. Monserrat, C. J. Pickard, and V. V. Brazhkin, “Speed of sound from fundamental physical constants,” *Sci. Adv.* **6** (2020), eabc8662.
- [18] T. Tao, “Finite time blowup for an averaged three-dimensional Navier-Stokes equation,” *J. Amer. Math. Soc.* **29**(3), 601–674, 2016.
- [19] L. Caffarelli, R. Kohn, L. Nirenberg, “Partial regularity of suitable weak solutions of the Navier-Stokes equations,” *Comm. Pure Appl. Math.* **35**(6), 771–831, 1982.
- [20] J. Leray, “Sur le mouvement d’un liquide visqueux emplissant l’espace,” *Acta Math.* **63**, 193–248, 1934.
- [21] P. Constantin, C. Foias, *Navier-Stokes Equations*, University of Chicago Press, 1988.
- [22] A. N. Kolmogorov, “The local structure of turbulence in incompressible viscous fluid for very large Reynolds numbers,” *Dokl. Akad. Nauk SSSR* **30**, 299–303, 1941.
- [23] M. J. Lighthill, “On sound generated aerodynamically. I. General theory,” *Proc. R. Soc. Lond. A* **211**(1107), 564–587, 1952.
- [24] I. Proudman, “The generation of noise by isotropic turbulence,” *Proc. R. Soc. Lond. A* **214**(1116), 119–132, 1952.
- [25] L. D. Landau, E. M. Lifshitz, *Fluid Mechanics* (Volume 6 of Course of Theoretical Physics, 2nd ed.), Pergamon Press, 1987.
- [26] G. K. Batchelor, *An Introduction to Fluid Dynamics*, Cambridge University Press, 1967.
- [27] D. Kleckner, W. T. M. Irvine, “Creation and dynamics of knotted vortices,” *Nature Physics* **9**(4), 253–258, 2013.
- [28] J. Schumacher, J. D. Scheel, D. Krasnov, D. A. Donzis, V. Yakhot, K. R. Sreenivasan, “Small-scale universality in fluid turbulence,” *Proc. Natl. Acad. Sci. USA* **111**(30), 10961–10965, 2014.
- [29] D. A. Donzis, P. K. Yeung, K. R. Sreenivasan, “Dissipation and enstrophy in isotropic turbulence: resolution effects and scaling in direct numerical simulations,” *Phys. Fluids* **20**, 045108, 2008.
- [30] G. Luo, T. Y. Hou, “Toward the finite-time blowup of the 3D axisymmetric Euler equations: a numerical investigation,” *Multiscale Model. Simul.* **12**(4), 1722–1776, 2014.
- [31] T. M. Elgindi, “Finite-time singularity formation for $C^{1,\alpha}$ solutions to the incompressible Euler equations on $\mathbb{R}^3$,” *Ann. of Math.* **194**(3), 647–727, 2021.
- [32] J. Chen, T. Y. Hou, “Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data,” arXiv:2210.07191, 2022.
- [33] Y. Wang et al., “Discovery of unstable singularities,” arXiv:2509.14185, 2025.
- [34] D. Albritton, E. Brué, M. Colombo, “Non-uniqueness of Leray solutions of the forced Navier–Stokes equations,” *Ann. of Math.* **196**(1), 415–455, 2022.
- [35] C. Foias, D. D. Holm, E. S. Titi, “The Navier–Stokes-alpha model of fluid turbulence,” *Physica D* **152–153**, 505–519, 2001.
- [36] D. D. Holm, J. E. Marsden, T. S. Ratiu, “Euler–Poincaré models of ideal fluids with nonlinear dispersion,” *Phys. Rev. Lett.* **80**, 4173–4176, 1998.

- [37] F. Golse, L. Saint-Raymond, “The Navier–Stokes limit of the Boltzmann equation for bounded collision kernels,” *Invent. Math.* **155**, 81–161, 2004.
- [38] A. V. Bobylev, “The Chapman–Enskog and Grad methods for solving the Boltzmann equation,” *Sov. Phys. Dokl.* **27**, 29–31, 1982.
- [39] P. L. Bhatnagar, E. P. Gross, M. Krook, “A model for collision processes in gases. I. Small amplitude processes in charged and neutral one-component systems,” *Phys. Rev.* **94**, 511–525, 1954.
- [40] A. Cheskidov, D. D. Holm, E. Olson, E. S. Titi, “On a Leray- α model of turbulence,” *Proc. R. Soc. A* **461**, 629–649, 2005.
- [41] D. Bandak, N. Goldenfeld, A. A. Mailybaev, G. Eyink, “Dissipation-range fluid turbulence and thermal noise,” *Phys. Rev. E* **105**, 065113, 2022.
- [42] P. Constantin, C. Fefferman, “Direction of vorticity and the problem of global regularity for the Navier–Stokes equations,” *Indiana Univ. Math. J.* **42**(3), 775–789, 1993.
- [43] Q. Zheng, D. J. Durben, G. H. Wolf, C. A. Angell, “Liquids at large negative pressures: water at the homogeneous nucleation limit,” *Science* **254**, 829–832, 1991.
- [44] F. Caupin, E. Herbert, “Cavitation in water: a review,” *C. R. Physique* **7**, 1000–1017, 2006.
- [45] T. Buckmaster, V. Vicol, “Nonuniqueness of weak solutions to the Navier-Stokes equation,” *Ann. of Math.* **189**(1), 101–144, 2019.
- [46] A. Bedroya, C. Vafa, “Trans-Planckian censorship and the swampland,” *J. High Energ. Phys.* **2020**(9), 123, 2020.